

The Codex.
IAQ’s business, easy to read.

Three sections for three readers. Section 1 is for anyone. Section 2 is for engineers and buyers who know the industry. Section 3 is how the website shows it.

IAQ member portal · the Codex · printed 27 October 2026

How to read every picture: the six kinds of thing.

One colour and one icon per kind, the same everywhere on this page. Summary tells them apart; Detailed says what each one is, what IAQ calls it, and what it is not.

Business unit

Three

The company inside IAQ that signs the contract and answers for the result. A buyer chooses the unit first, because the unit decides who is accountable.

- **EPC**
  Engineering, Procurement and Construction. Builds the facility.
- **PCU & TTI**
  Process Critical Utilities & Total Tool Installation. Re-equips a live semiconductor fab.
- **EFM**
  Energy Facility Management. Runs and maintains it, and cuts the energy bill.

"We bought the whole fab from the EPC unit. Two years later EFM took over the chiller plant."

Think The company you sign with.

Not Not a service. A unit sells several services; "EPC" is a unit and also the name of a contract model, which is why it gets confused.

On the site The first group of the Services menu, and the three unit pages.

Delivery model

Four

How the contract runs: who holds the construction contracts, who answers for the system performing, and how IAQ is paid. Every model belongs to one unit.

- **EPCC**
  IAQ delivers everything, commissioning included, and answers for the result. EPC unit.
- **EPCM**
  The owner holds the contracts, IAQ manages them. EPC unit.
- **Cooling as a Service**
  IAQ funds and runs the chiller plant for a fixed tariff. EFM.
- **Energy Performance Contracting**
  IAQ funds the upgrade and is paid from the measured savings. EFM.

"Same six services, but on EPCM the owner keeps the contracts and IAQ manages the programme."

Think The shape of the deal.

Not Not a service and not a unit. It changes who is accountable, not what gets built.

On the site Grey chips under each unit in the Services menu; the four-ways slide; each unit page.

Service

Six

What is written into the contract, in the order a project runs. Any one can be bought alone, or several, or all six as one cycle.

- **1 Design**
  Engineering design and consultation.
- **2 Procure**
  Procurement.
- **3 Construct**
  Construction.
- **4 Commission**
  Testing and commissioning.
- **5 Tool Installation**
  Tool Installation: production tools connected and released.
- **6 Maintain**
  Maintenance.

"Design only, please. We already hold the rest of the scope."

Think The menu.

Not Not a system. Process cooling water is a system that two services touch; it is never a service itself.

On the site The six-service strip in the Services menu, the delivery cycle, and one page per service.

Work

Three

The discipline that does the work on site, which is how the market classifies a contractor. Most contractors carry one; IAQ carries all three in house.

CSA
Civil, Structural and Architectural: the building itself.

MEP
Mechanical, Electrical and Plumbing: what makes it run.

Process utilities
The utilities a production tool consumes: gases, chemicals, pure water, exhaust.

"A civil contractor does CSA only. IAQ does CSA, MEP and process, so the whole scope stays with one firm."

Think The trades on site.

Not Not a service. A service is what you buy; the work is who does it and how they are classified.

On the site The work layer on every unit page, and the three-disciplines slide.

System

Thirteen listed

The physical thing being built, always inside one kind of work. A system is what the drawings show and what commissioning proves.

Process cooling water
MEP.

Ultrapure water
Process utilities.

Fan filter units
MEP.

Fire protection
MEP.

Specialty gases
Process utilities.

Cleanroom envelope
CSA.

"The PCW loop is built in MEP work, then the last metres to each tool are hookup."

Think The parts of the fab.

Not Not a service, even when it is priced on its own. "Buy the PCW package" is a scope of construction, not a seventh service.

On the site Green chips under each kind of work on the unit pages; the fab model on the home page.

Market

Seven

Who the facility is for. Each market measures a facility by something different, so it decides the class or spec IAQ builds to.

Semiconductor
Particle count, ISO 3 to 7.

Data centre
Uptime and thermal stability.

EV battery
Dew point, dry room.

Photovoltaics
Process stability at scale.

District cooling
Kilowatt hours.

Bio LifeScience
GMP grade.

Food & Beverage
Hygiene regime.

"An EV battery plant is judged by dew point, so the dry room is the spec that matters."

Think The customer.

Not Not a unit and not a service. IAQ serves every market with the same three units.

On the site The Markets menu, the home page market rail, one page per market.

Side by side, what each one really is.

Compared	Business unit	Delivery model	Service	Work	System	Market
What it is	The company inside IAQ that signs the contract and answers for the result. A buyer chooses the unit first, because the unit decides who is accountable.	How the contract runs: who holds the construction contracts, who answers for the system performing, and how IAQ is paid. Every model belongs to one unit.	What is written into the contract, in the order a project runs. Any one can be bought alone, or several, or all six as one cycle.	The discipline that does the work on site, which is how the market classifies a contractor. Most contractors carry one; IAQ carries all three in house.	The physical thing being built, always inside one kind of work. A system is what the drawings show and what commissioning proves.	Who the facility is for. Each market measures a facility by something different, so it decides the class or spec IAQ builds to.
How many IAQ has	Three	Four	Six	Three	Thirteen listed	Seven
The names	EPC · PCU & TTI · EFM	EPCC · EPCM · Cooling as a Service · Energy Performance Contracting	1 Design · 2 Procure · 3 Construct · 4 Commission · 5 Tool Installation · 6 Maintain	CSA · MEP · Process utilities	Process cooling water · Ultrapure water · Fan filter units · Fire protection · Specialty gases · Cleanroom envelope	Semiconductor · Data centre · EV battery · Photovoltaics · District cooling · Bio LifeScience · Food & Beverage
Who decides it	The buyer, first of all.	The buyer, with IAQ, before the contract is written.	The buyer, per contract.	The market, when it classifies a contractor.	The design, then the commissioning test.	Who is paying for the facility.
Think of it as	The company you sign with.	The shape of the deal.	The menu.	The trades on site.	The parts of the fab.	The customer.
Often confused with	Not a service. A unit sells several services; “EPC” is a unit and also the name of a contract model, which is why it gets confused.	Not a service and not a unit. It changes who is accountable, not what gets built.	Not a system. Process cooling water is a system that two services touch; it is never a service itself.	Not a service. A service is what you buy; the work is who does it and how they are classified.	Not a service, even when it is priced on its own. “Buy the PCW package” is a scope of construction, not a seventh service.	Not a unit and not a service. IAQ serves every market with the same three units.
Where you see it on the site	The first group of the Services menu, and the three unit pages.	Grey chips under each unit in the Services menu; the four-ways slide; each unit page.	The six-service strip in the Services menu, the delivery cycle, and one page per service.	The work layer on every unit page, and the three-disciplines slide.	Green chips under each kind of work on the unit pages; the fab model on the home page.	The Markets menu, the home page market rail, one page per market.
Colour in every picture	Blue, filled	Ink outline	Red	Amber	Green	Violet

Ten questions to ask IAQ, with our answer on each.

Nothing waits on them. Under each question: what was done meanwhile, and the answer Brand Method would give if the call were ours.

1 **Question for Nabilah**

Unit 2 is now “Process Critical Utilities & Total Tool Installation Solutions”, IAQ’s own written name, in place of “Total Tools Hookup Solution”. Confirm?

Meanwhile Applied on the site and the slides on 18 Sep, from the cross-check. Hookup now names service 6 only.

Our answer Confirm it. It is IAQ’s own written name from the unit questionnaire, and with it “hookup” means one thing only, service 6. Two names for one unit was the single most confusing thing on the old site.

2 **Question for Nabilah**

IAQ’s papers still say founded 1994 and eight countries; the site says 1995 and seven, on your instruction of 10 Sep. Which is final?

Meanwhile The site keeps 1995 and seven.

Our answer Go by the incorporation certificate, and correct whichever side is wrong, including IAQ’s own website. Until it is checked the site keeps 1995 and seven, because the 10 Sep instruction is the latest word. Count countries by offices, not by projects built, and say “projects in more” where it matters.

3 **Question for Nabilah**

Do the six service pages stay as depth reached from the cycle, or fold into the Services page?

Meanwhile Kept, out of the menu, reached from the cycle diagram only.

Our answer Keep them. Each one is a landing page for a search such as “cleanroom commissioning Malaysia”, and a buyer who wants one service wants one page. Out of the menu is right: the cycle and the hub are the way in.

4 **Question for The unit heads**

Which of the six services does each unit carry? Please confirm the table on the six services slide, most of all the ones marked “when asked”.

Meanwhile EPC **(Engineering, Procurement and Construction)** carries the first four: design, procurement, construction and commissioning. The hookup unit carries five, and maintenance when asked. EFM **(Energy Facility Management)** carries maintenance and commissioning, and the first three when an upgrade needs them. Read from the review and the questionnaires, not yet confirmed by the unit heads.

Our answer Confirm the table as drawn. It follows the 17 Sep review and the questionnaires word for word, and the only cells with any doubt are the “when asked” ones, which cost nothing if a unit head removes one.

5 **Question for Nabilah, the BIM** (Building Information Modelling) **manager**

Where do cleanroom systems sit in the work layer?

Meanwhile The envelope (walls, ceiling grid, raised floor) sits under CSA, and the fan filter units under MEP.

Our answer Envelope under CSA (Civil, Structural and Architectural), fan filter units under MEP (Mechanical, Electrical and Plumbing). That is how the Revit model is coded and how tender packages split, so the site should not invent a third home for them.

6 **Question for Shazwan, Nabilah**

What page size does the company profile use, so the slides are placed at the right ratio?

Meanwhile The set exports at 16:9, 1920 by 1080, as PNG and as one PDF. It reflows to A4 landscape with a margin; a portrait profile needs a second layout.

Our answer 16:9 landscape, 1920 by 1080. The set exports natively at that size, reads on a screen, prints on A4 landscape with a margin, and a portrait profile would mean a second layout of every slide for no gain.

7 **Question for Nabilah, management**

Which figures will IAQ release for a performance slide like the benchmark’s: headcount by function, the safety rate, order intake?

Meanwhile Only the six counters the site already publishes are used. Revenue stays off, as on the site.

Our answer The six counters the site already publishes, plus safe manhours, which IAQ publishes itself. Revenue and order intake stay off until listing counsel clears them; a number that later has to come down does more harm than a slide without it.

8 **Question for Bazil**

Do slides 2 and 5 go onto the public Services page in place of the unit cards and the cycle?

Meanwhile They live in the Codex and export for the profile. The public page keeps its current sections until a responsive version is built.

Our answer Yes, as a responsive rebuild of what they show, not as pictures: the three units with what each carries, the one-picture map and the reading order. That is the Services page as built on 24 Sep; the slides themselves are 16:9 images and belong in the profile.

9 **Question for Sunny**

When does the approved project list arrive?

Meanwhile The proof slide is computed from the 18 published references and grows on its own as the registry does.

Our answer Ask Sunny for it in the description format on the project format page, one project at a time if that is faster. Until then the proof slide computes from the 18 published references, so nothing stalls.

10 **Question for Nabilah**

When does IAQ supply the capability statement PDF?

Meanwhile The Downloads tab in the member portal holds the button, inert until the file arrives. The three ISO certificates download now.

Our answer Send the current PDF even if it is dated. The portal button goes live the day it lands and the file can be swapped at any time; the button staying inert is what a member notices.

1

For everyone. What IAQ does, in plain words.

IAQ is one company with three businesses. Together they build a high-tech factory, fit it out and keep it running. No industry words needed.

1

It builds the factory.

From the first drawing to a finished, clean facility.

EPC (Engineering, Procurement and Construction)

2

It connects the machines.

When production machines arrive in a chip factory, it brings them their gases, chemicals, water and power.

PCU & TTI (Process Critical Utilities & Total Tool Installation)

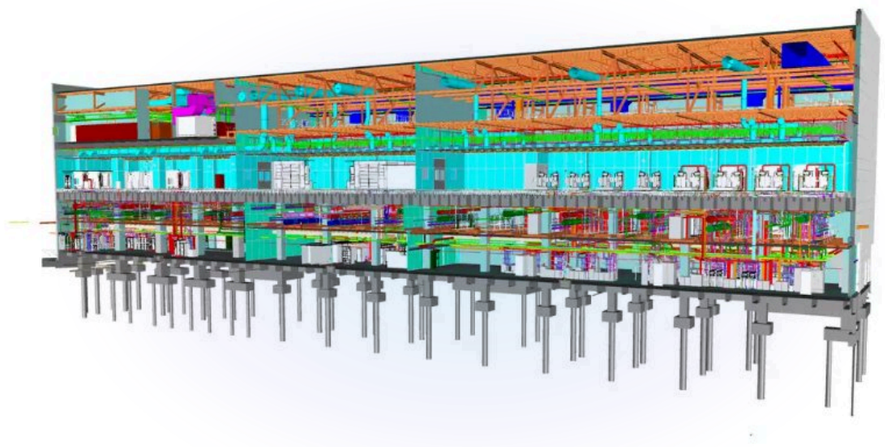
3

It runs it for less.

It maintains the plant and cuts the energy bill, funded by IAQ upfront.

EFM (Energy Facility Management)

Watch a fab being built, and who does each part



1. The ground and the frame Piles go into the ground. The structure, the floors and the roof go up. **EPC (Engineering, Procurement and Construction)** builds the facility. Work: CSA (Civil, Structural and Architectural). Services: Engineering design, Procurement, Construction.

2. Air, water and power Ducts, pipes and cables run through every level, so the building can breathe, cool and be powered. **EPC (Engineering, Procurement and Construction)** builds the facility. Work: MEP (Mechanical, Electrical and Plumbing). Services: Construction.

3. The clean room Sealed walls, a ceiling of air filters and a raised floor keep the air far cleaner than the air outside. **EPC (Engineering, Procurement and Construction)** builds the facility. Work: CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing). Services: Construction, Testing and commissioning.

4. The machines arrive The production machines are moved in. Each one is connected to its gases, chemicals, pure water and power, then tested. **PCU & TTI (Process Critical Utilities & Total Tool Installation)** connects the machines. Work: Process utilities, MEP (Mechanical, Electrical and Plumbing). Services: Tools hookup, Testing and commissioning.

5. Running for years The fab runs day and night. It is maintained, and its energy bill is brought down with no money paid upfront. **EFM (Energy Facility Management)** runs it and cuts the energy bill. Work: MEP (Mechanical, Electrical and Plumbing). Services: Maintenance.

The life of a facility, and who to call at each point

1

Design and build

12 to 24 months

From the first drawing to handover, one party accountable.

Call EPC (Engineering, Procurement and Construction) **when**

You are building or extending a facility and want one company to answer for all of it.

2

Tools in, fab live

Per tool set

Utilities to each tool, inside a fab that keeps running.

Call PCU & TTI (Process Critical Utilities & Total Tool Installation) **when**

Your fab is running and new tools are coming in, or the gas, chemical, water or exhaust systems need to grow.

3

Run and maintain

5 to 20 years

The plant run under a tariff, or the upgrade paid from savings.

Call EFM (Energy Facility Management) **when**

Your energy or cooling bill is high and you want it cut, with IAQ funding the work and paid from the saving.

The six services, one after another



Design 1



Procurement 2



Construction 3



Commissioning 4

Tool Installation 5

Maintenance 6

How a district cooling system works



Questions clients ask, answered

Choosing a model

How IAQ is engaged, and who answers for what.

4

Which model should I choose?

EPCC when you want one party to deliver everything and answer for the result. EPCM when you hold the construction contracts yourself and want IAQ to manage them. Large multinationals engage EPCM on the biggest programmes.

See EPC →

What is the difference between EPCC and EPCM?

EPCC is EPC plus Commissioning: IAQ delivers the whole project, commissioning included. EPCM is EPC Management: IAQ manages the project to the same standard, and the owner holds the construction contracts.

Can I buy one service on its own?

Yes. Procurement only, design only, or design, procure and construct. The scope changes; the EPC model still governs how it runs.

Who answers for a system's performance?

Under EPCC, IAQ: one contract, one party, from the first drawing to a facility proven to operate as intended. Under EPCM the owner holds the contracts and IAQ manages them; the standard the work is held to is the same.

Time and cost

How long each engagement runs, and what is paid when.

4

How long does a facility take?

Typically 12 to 24 months under EPC, depending on the project. Tool hookup is priced and scheduled per tool set.

How is tools hookup priced?

Per tool set. Each set is scoped from the tool maker's facility requirements, so the price follows the tools you bring in, not the size of the fab.

See PCU & TTI →

What does an energy contract cost up front?

Nothing. Under Cooling as a Service IAQ funds, builds and runs the chiller plant for a fixed tariff over 10 to 20 years. Under Energy Performance Contracting IAQ funds the upgrade and is paid from the measured savings over 5 to 10 years.

See EFM →

How are the savings measured?

Against an audited baseline agreed before the work starts. IAQ is paid from the measured saving, so the saving has to be real and has to last.

A live fab

Bringing tools into a fab that keeps running.

4

My fab is live. Can tools be brought in while it keeps running?

That is the PCU & TTI unit. It works in four phases, facilitization, rig-in, hook-up and commissioning, and it carries all six services except maintenance, which it adds when asked.

See PCU & TTI →

What are the four phases of a tool installation?

Facilitization, before the tool arrives: everything upstream of its connection point is built to the tool maker's requirements. Rig-in: the tool is uncrated in a de-trash airlock, moved along an agreed route, set, levelled and anchored. Hook-up: each utility is tapped from the facility and connected into the tool, to its own standard. Commissioning: the tool is proven, then released to production.

What do Process Critical Utilities cover?

The systems a production tool runs on: clean dry air, specialty gases and chemical delivery, ultrapure water, process vacuum and exhaust, and waste treatment.

Does PCU & TTI work outside semiconductor?

No. Semiconductor only. Other hi-tech facilities are built and equipped by EPC under the same six services.

After handover

Running and maintaining what was built.

3

What happens after handover?

Maintenance is the fifth service and EFM carries it as its core. What the maintenance team learns goes back into the next design.

See Maintain →

What does maintenance cover?

Planned care of the plant and systems that keeps the facility at the standard it was handed over at: uptime protected, lifespan protected, records kept.

Does IAQ take on facilities built by others?

Yes. Energy contracts are for facilities already running, and the tools unit is bought on a fab built by IAQ or by others.

Working with IAQ

Standards, reach, and how to start.

4

What standards is IAQ certified to?

ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018, and CIDB Grade G7 in Malaysia.

About the Group →

Where does IAQ work?

Headquartered in Shah Alam, Malaysia, with Penang, Singapore, Germany and India among its offices, and project teams across Asia and Europe. Delivering since 1995.

Global Presence →

Is the design done in BIM?

Yes. Every discipline is designed in one coordinated 3D model, so clashes are found on screen, not on site. The fab on the cover of this page is IAQ's own model.

How do I start?

Tell us what you are planning, or what is running and needs to change.

Start a project →

Technical: EPCC and EPCM

The two contract models side by side, point by point.

5

Under EPCC and EPCM, who holds the construction contracts?

Under EPCC, IAQ holds them. Under EPCM, the owner holds the individual construction contracts and IAQ manages them on the owner's behalf.

Who answers for the system performing as intended?

Under EPCC, IAQ answers for it, as the one accountable party. Under EPCM, IAQ manages the project and the owner's contractors deliver the works.

Is commissioning and start-up included?

Under EPCC it is included in IAQ's scope. Under EPCM it is coordinated by IAQ.

How long is a typical EPC programme?

Typically 12 to 24 months, under either model, depending on the project.

When is EPCC the better choice, and when is EPCM?

EPCC when you want one accountable party for the whole facility. EPCM when you want direct control of the construction contracts, with IAQ running coordination, programme and cost.

Cover

Cover · One fab, every layer
Everything inside a fab, layer by layer, and which kind of work delivers each layer.

Profile: opens the business section. Website: the Services page hero. Booth screen, beside the 3D model.

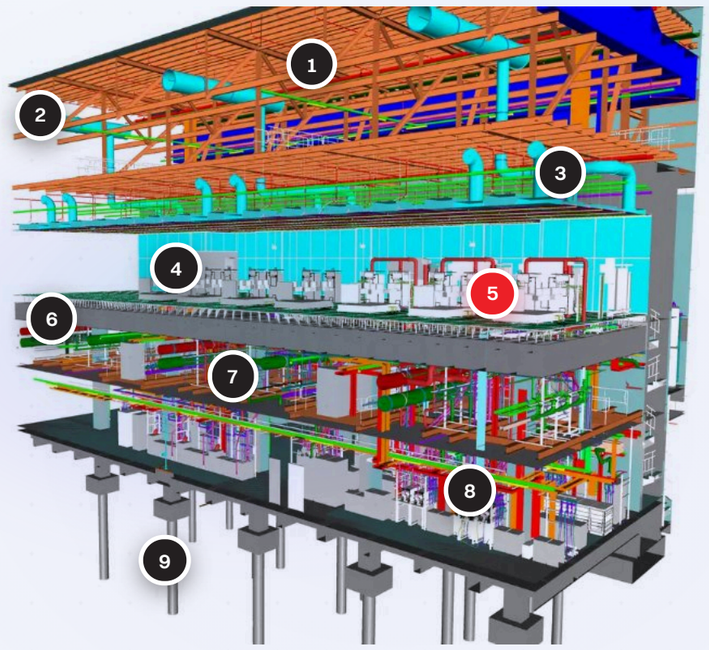
One fab, every layer: what IAQ delivers.

A section through IAQ's own Revit coordination model. Every numbered layer is designed, procured, built and proven by the same company.

CSA (Civil, Structural and Architectural)

The building

1. Roof steel and structural frame
4. Cleanroom envelope
6. Waffle slab and raised floor
9. Piles and pile caps



MEP (Mechanical, Electrical and Plumbing)

What makes it run

2. Air handling and ducting
3. Building services, the interstitial level
8. Sub-fab plant and fire protection

Process utilities

What the tools run on

7. Process utilities in the sub-fab

Tools hookup

Service 6, the tools themselves

5. Process tools, hooked up

It assembles in the order IAQ builds it

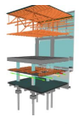
The same model, six frames of sixty one



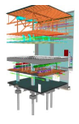
1 Piles and pile caps



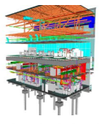
2 Frame, slabs and roof steel



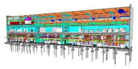
3 Cleanroom envelope



4 Building services



5 Tools and sub-fab plant



6 The whole fab

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · BIM (Building Information Modelling)

IAQ at a glance: one group, three business units.

A total facility solutions provider for controlled environments. Each unit answers one question a facility owner has.

Since 1995
31 years, founded in Malaysia as a cleanroom specialist

450 people
Engineers, project teams and site crews, group wide

250+ projects
Delivered for hi-tech manufacturers and facility owners

1,050,000 m²
Of cleanroom built

7 countries
Eight offices, headquartered in Shah Alam

ISO 3
The cleanest class IAQ has built



Business unit 1

EPC (Engineering, Procurement and Construction)

Builds the facility.

- Design, procurement, construction and commissioning
- Cleanrooms, dry rooms and hi-tech facilities
- CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) and process utilities, in house

EPCC · EPCM



Business unit 2

Process Critical Utilities & Total Tool Installation Solutions

Re-equips a live semiconductor fab.

- Utilities: gas, chemical and slurry, water, process exhaust
- Tool installation: facilitize, rig in, hook up, commission
- Semiconductor fabs, front end to back end

Standalone · EPC or EPCM model



Business unit 3

EFM

Runs and maintains it.

- Energy Facility Management, a registered ESCO (Energy Service Company)
- IAQ funds the upgrade, paid from the savings
- Chiller plant and district cooling, run under a service level

Cooling as a Service
Energy Performance Contracting

Seven markets

Semiconductor · Data Centre · EV Battery · Photovoltaics · District Cooling & Heating · Bio LifeScience · Food & Beverage

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management)

Three units, across the life of your facility.

Read it as the owner does: left to right, from the first drawing to a plant in operation, and round again at the next expansion.

Business unit 1
EPC
Builds the facility.

EPCC · IAQ delivers | EPCM · IAQ manages

Business unit 2
PCU & TTI
Process Critical Utilities & Total Tool Installation

Business unit 3
EFM
Runs and maintains it.

Can also sit inside the same EPC contract

1 Plan and design	2 Procure	3 Build	4 Commission and hand over	5 Tools move in	6 Operate and maintain
The service	The service	The service	The service	The service	The service
Engineering design and consultation	**Procurement**	**Construction**	**Testing and commissioning**	**Tools hookup**	**Maintenance**
You receive	You receive	You receive	You receive	You receive	You receive
Drawings, the BIM model and the permits	Equipment on site, on programme and on budget	The facility, built to its class	Proof it performs, and the certified documents	Tools connected, qualified and released to production	Uptime, compliance and a lower energy bill

And round again. New tools, a new line or an energy upgrade start the next design. The same three units carry it.

Short forms: EPC (Engineering, Procurement and Construction) · EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · EFM (Energy Facility Management) · BIM (Building Information Modelling)

Watch it: three short films

New to cleanrooms and chip factories? Start here. Nothing loads until you press play.

1. **What a cleanroom is** (Lighthouse Worldwide Solutions, 2:25): youtube.com/watch?v=vSS_uJIrvvk
2. **A tour of a TSMC fab** (TSMC Museum of Innovation, 7:55): youtube.com/watch?v=divOKxuYkIM
3. **Bosch opens its Penang chip plant** (Bosch Malaysia, 1:28): youtube.com/watch?v=VUG1QFOCL2E

2

For professionals. How the industry reads IAQ.

Engineers and buyers in hi-tech construction do not shop for a service. They ask three questions, in this order, and classify a contractor by its answers.

1 Which model?

Model

How the job is contracted. EPCC (Engineering, Procurement, Construction and Commissioning): IAQ delivers everything and answers for the result. EPCM (Engineering, Procurement and Construction Management): the owner holds the contracts and IAQ manages them. A single service on its own. Or, for energy, an ESCO (Energy Service Company) contract paid from the savings.

2 Which scope?

Service

What the contract covers. One of the six services, several, or all six. For example procurement only, design only, or design, procure and build.

3 Which work?

Work

Which kind of work the contractor does, which is how the market classifies it: CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) or process utilities. Most contractors do one. IAQ does all three in house.

Everything IAQ does, and how it connects

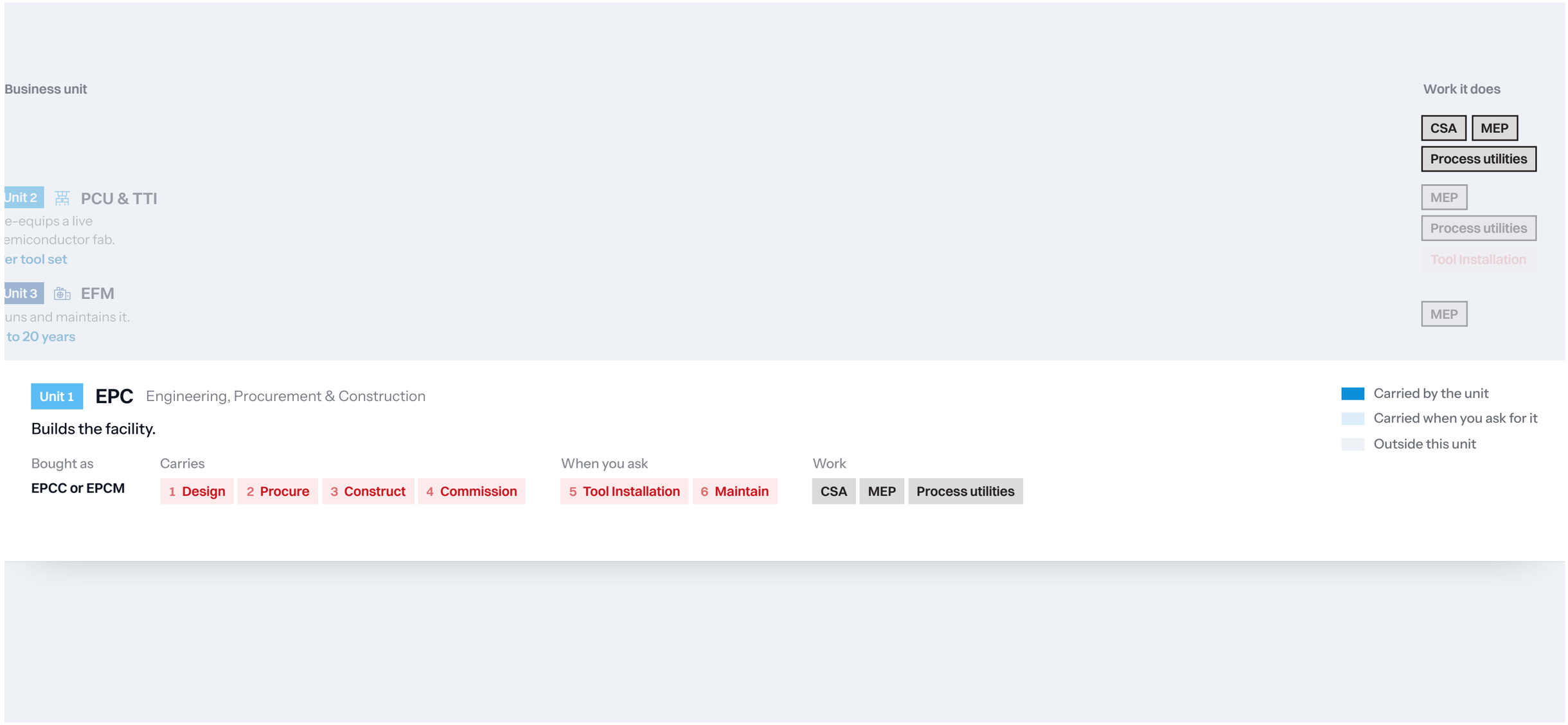
Pick any service, unit, discipline or system. Its connections light up, and one sentence says what they mean.

Business unit	Bought as	Services it carries	Only when the client asks	Its work
EPC (Engineering, Procurement & Construction)	EPCC (Engineering, Procurement, Construction and Commissioning) or EPCM (Engineering, Procurement and Construction Management)	Engineering design and consultation, Procurement, Construction and Testing and commissioning	Tool Installation and Maintenance	CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) and Process utilities
Process Critical Utilities & Total Tool Installation Solutions (PCU & TTI)	A standalone job, or inside an EPC (Engineering, Procurement and Construction) or EPCM (Engineering, Procurement and Construction Management) contract	Engineering design and consultation, Procurement, Construction, Testing and commissioning and Tool Installation	Maintenance	MEP (Mechanical, Electrical and Plumbing), Process utilities and Tool Installation
EFM (Energy Facility Management)	Cooling as a Service, Energy Performance Contracting or Build Operate Transfer	Maintenance and Testing and commissioning	Engineering design and consultation, Procurement and Construction	MEP (Mechanical, Electrical and Plumbing)

Work	The systems inside it
CSA (Civil, Structural and Architectural)	Piling and foundations, Structural frame and roof, Cleanroom envelope and Architectural finishes
MEP (Mechanical, Electrical and Plumbing)	Air conditioning and ventilation (HVAC, ACMV), fan filter units, Process cooling water (PCW), Chiller plant and district cooling, Fire protection, Electrical, high and low voltage and Plumbing and drainage
Process utilities	Clean dry air (CDA), Specialty gases and chemical delivery, Ultrapure water (UPW), Process vacuum (PV) and exhaust and Waste treatment
Tool Installation	Gas, chemical, water and exhaust connections to each tool, Power and controls to each tool and Release to production

The same, on one chart

The chart under the cover of the Services page: which unit carries which service, the work each one does, and where that work sits in the fab.



Tool Installation

Service 5, the tools themselves

Each production tool connected and released to production.

Done by PCU & TTI

In the model 5

The four kinds of work, and the systems in each

The three disciplines and the tools themselves, with the systems each covers, the units that do it and where it sits in the model.

Why tools hookup is one of the four works

Hookup is service 6 in the cycle, and a trade of its own. CSA, MEP and process utilities build the facility; hookup works at each production tool, joining it to those utilities.

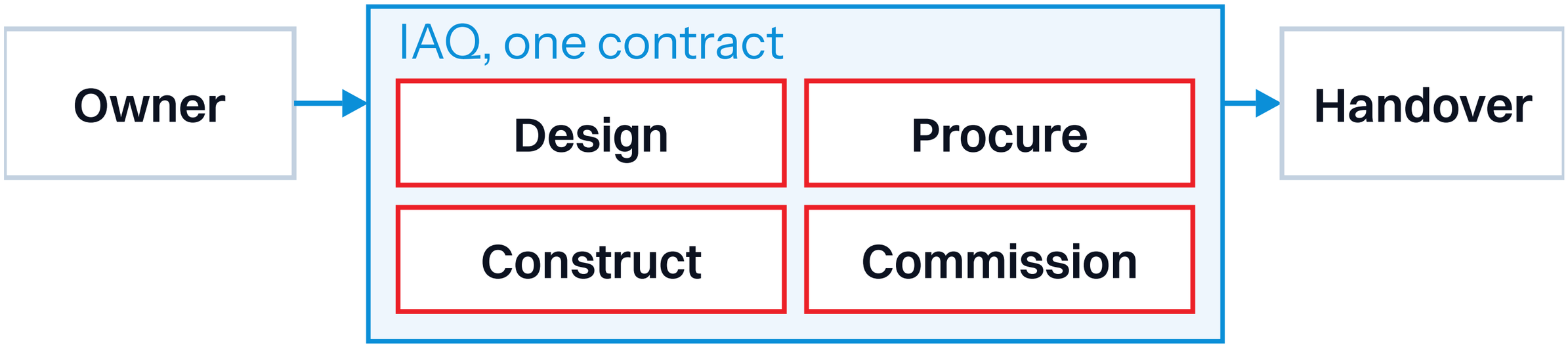
How each business unit works

The three drawings from the units board on the Services page: the flow of each unit, service red, unit blue, work yellow, systems green.

EPC

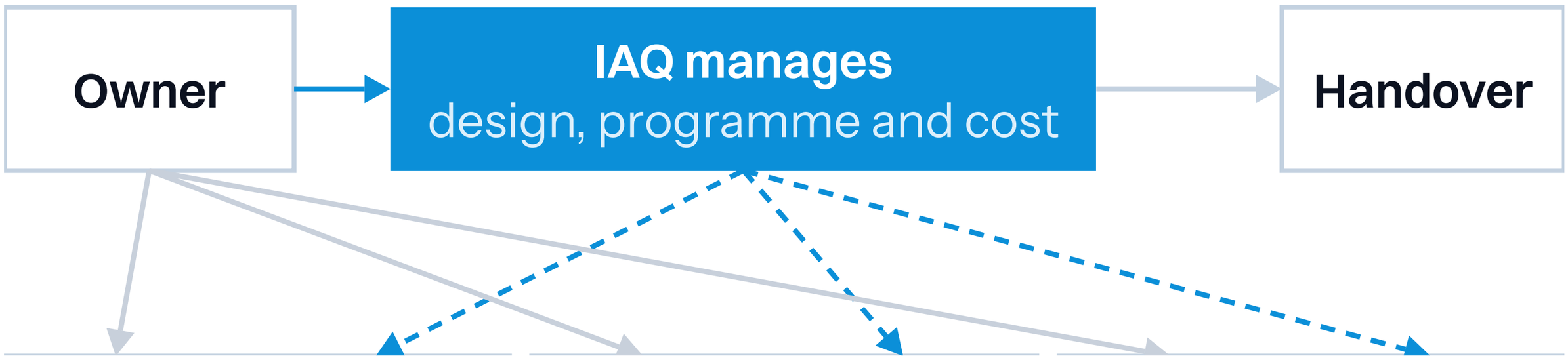
EPCC · one contract, IAQ delivers everything

IAQ answers for the system working as intended



EPCM · the owner holds the contracts

IAQ manages them; engaged on the biggest programmes



Contractor A
owner’s contract

Contractor B
owner’s contract

Contractor C
owner’s contract

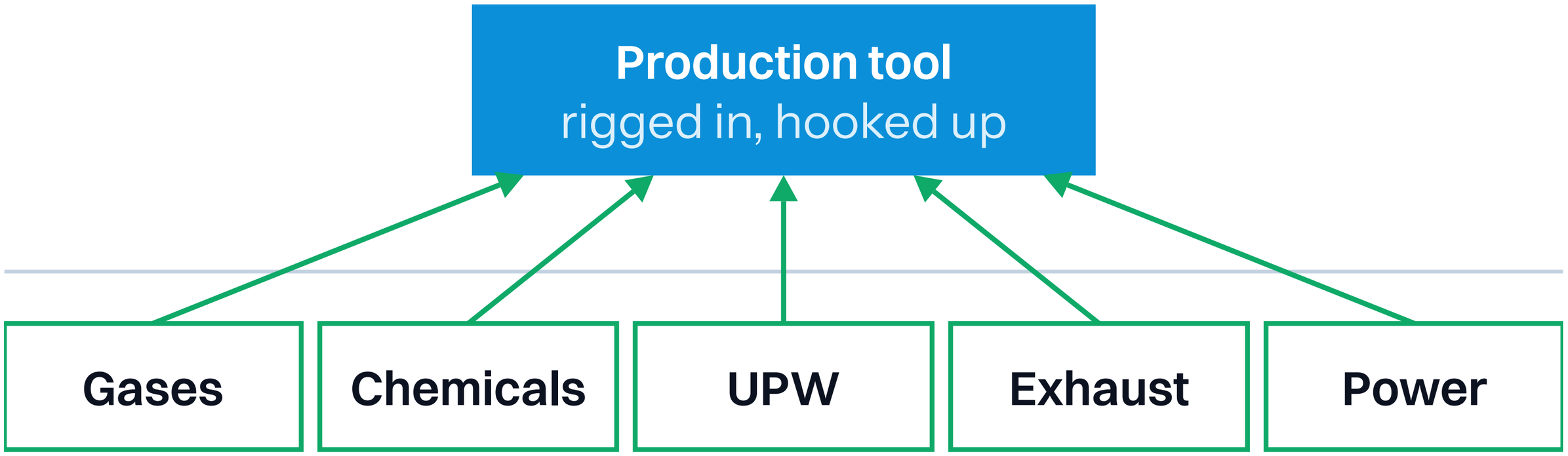
grey: the owner’s contract · blue dashed: IAQ manages that contractor

PCU & TTI

PCU builds the systems, TTI connects each tool

Inside a fab that keeps running. Semiconductor only.

Fab floor



Sub-fab · process utilities

IAQ's four phases, per tool

1 Facilitize
drops, power and room to spec

2 Rig in
moved in, set and levelled

3 Hook up
each utility tapped and connected

4 Commission
leak tested, purged, released

Cooling as a Service · IAQ owns the plant

IAQ's capital, the owner's saving

IAQ funds and builds
the chiller plant

IAQ operates it
10 to 20 years

Owner pays a tariff
per unit of cooling

Plant handed over
at the end

Energy Performance Contracting

The upgrade pays for itself, from the saving measured

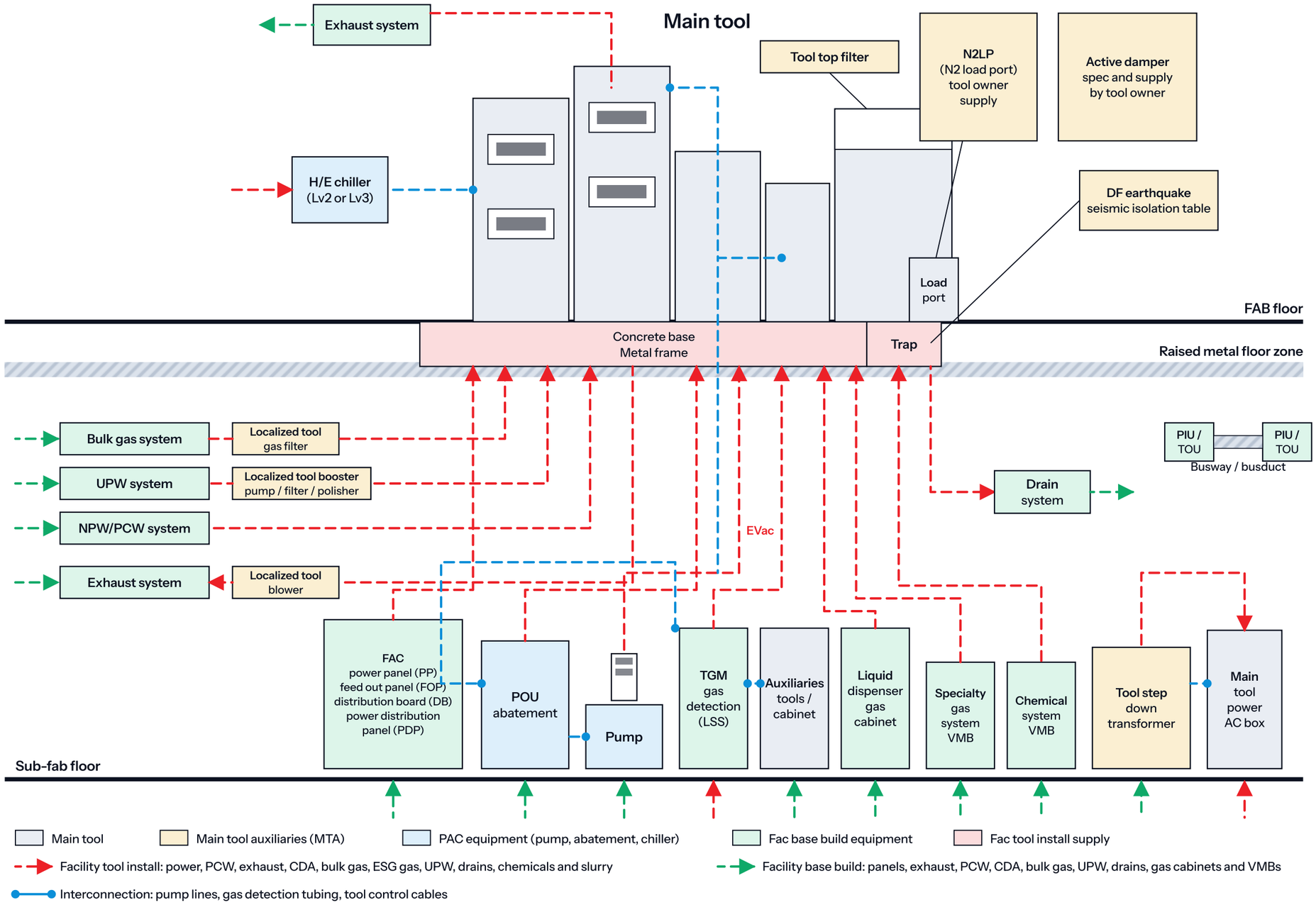
Audit
the baseline

IAQ funds
the upgrades

Savings measured
against baseline

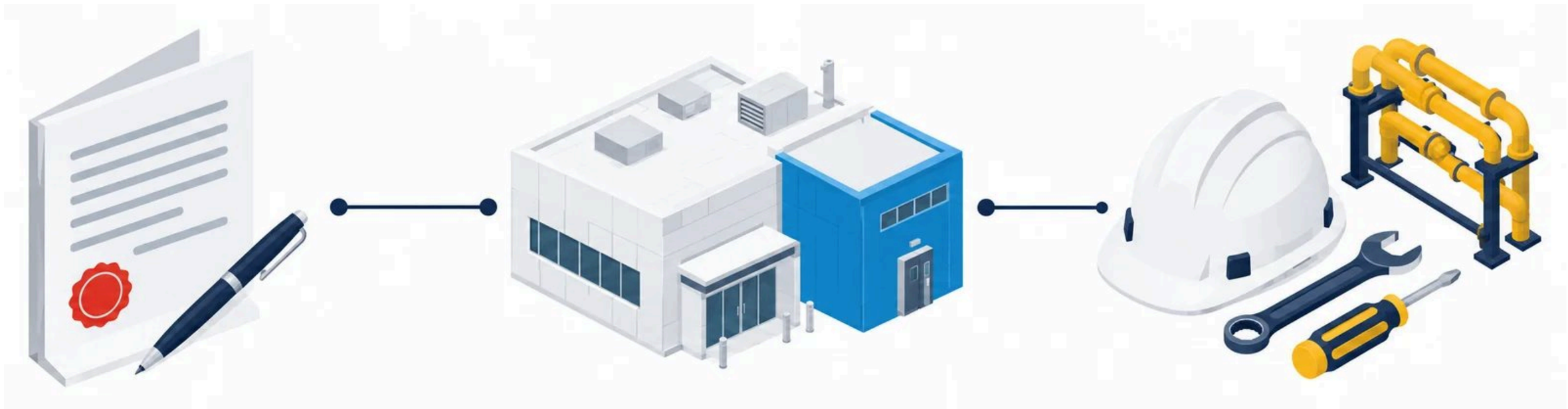
A tool hookup, as IAQ draws it

IAQ's own Main Tool schematic, redrawn: the tool on the fab floor, the raised floor, the sub-fab equipment, and the lines between them in the direction they run.



Three choices shape a quote

The model, the scope and the work, drawn as the Services page shows them.



Who does what, against the other kinds of contractor

The market classifies a contractor by the work it delivers. A process specialist does process only; a civil main contractor does CSA only; an M&E contractor does MEP only. IAQ delivers all five kinds, which is what "inclusive" means on this site.

Contractor	CSA	MEP	Process	Tool Installation	Energy, ESCO
IAQ Inclusive	■	■	■	■	■
A process specialist Process only			■		
A civil main contractor CSA only	■				
An M&E contractor MEP only		■			

3

IAQ in one picture

The six kinds of thing, one colour each: unit, model, service, work, system, market.

Profile: before the scope of services. Website: Services, beside the relationship map.

IAQ in one picture: which kind of thing is which.

Six kinds of thing, one colour each, the same on every slide. Read down: who you buy from, how the contract runs, what is in it, who does it, what gets built, and for whom.

Business unit

Who you buy from. Three.

- **EPC** — Engineering, Procurement and Construction. Builds the facility.
- **Process Critical Utilities & Total Tool Installation Solutions** — Re-equips a live semiconductor fab.
- **EFM** — Energy Facility Management. Runs and maintains it.

Delivery model

How the contract runs. Belongs to a unit.

- EPC: EPCC, EPCM
- Process Critical Utilities & Total Tool Installation Solutions: Standalone, On the EPC or EPCM model
- EFM: Cooling as a Service, Energy Performance Contracting

Service

What is in the contract. Six, one or all.

1. Engineering design
2. Procurement
3. Construction
4. Testing and commissioning
5. Tool Installation
6. Maintenance

Work

The discipline that does it. How the market classifies a contractor.

- **CSA** — Civil, Structural and Architectural
- **MEP** — Mechanical, Electrical and Plumbing
- **Process utilities** — The utilities a production tool consumes

System

The thing being built, inside a discipline.

- CSA: Piling and foundations, Structural frame and roof, Cleanroom envelope, Architectural finishes
- MEP: HVAC, ACMV, FFU, Process cooling water (PCW), Chiller plant, Fire protection
- Process utilities: Clean dry air (CDA), Specialty gases and chemical delivery, Ultrapure water (UPW), Process vacuum (PV) and exhaust

Market

Who it is for. Seven.

- Semiconductor
- Data Centre
- EV Battery
- Photovoltaics
- District Cooling & Heating
- Bio LifeScience
- Food & Beverage

Short forms: HVAC (heating, ventilation and air conditioning) · ACMV (air conditioning and mechanical ventilation) · VMB (valve manifold box) · M&E (mechanical and electrical)

Three words that mean two things.

Found by cross-checking IAQ's own documents against the 17 September review and the site. The colour says which kind of thing each meaning is.

EPC

One word, used for three different kinds of thing. This is where most of the confusion starts.

Business unit	**EPC** (Engineering, Procurement and Construction)**, unit 1** The unit that builds the facility.
Delivery model	**The EPC model: EPCC** (Engineering, Procurement, Construction and Commissioning)**, EPCM** (Engineering, Procurement and Construction Management) How a contract runs. The hookup unit works on it too.
A contract	**Energy Performance Contracting** EFM (Energy Facility Management)'s contract. Always written in full, never EPC.

Say it this way

Say "the EPC unit" or "the EPC model".
Never shorten the energy contract.

Hookup

Unit 2 went by two names, and one of them borrowed the name of a service.

Business unit	**Process Critical Utilities & Total Tool Installation Solutions** Unit 2, IAQ's own written name. "Total Tools Hookup Solution" is retired.
Service	**Tools hookup, service 6** The work of connecting a tool. The only thing "hookup" names now.

Say it this way

Name the unit by what it sells. Keep "hookup" for the work.

PCW, CDA, UPW

Systems turn up in two places, so they get listed as services by mistake.

System	**Built with the facility** PCW (process cooling water) is mechanical work. CDA (clean dry air) and UPW (ultrapure water) are process work.
Service	**Their last metres are hookup** From the main to the tool, service 6 connects them.

Say it this way

A system is never a service. It appears twice because two crews touch it.

Six services. Buy one, or all of them.

IAQ calls the six its total solution: the scope written into the contract, in the order a project runs. Under it, the business unit that carries each one.

	Before site		On site		In operation	
The service	1 **Engineering design and consultation**	2 **Procurement**	3 **Construction**	4 **Testing and commissioning**	5 **Tool Installation**	6 **Maintenance**
The scope	• Feasibility and master planning • Concept to detailed design, CSA (Civil, Structural and Architectural), MEP and process • BIM coordination • Permitting and authority liaison	• Supplier prequalification • Tenders, estimates and cost analysis • Long-lead equipment tracked	• Site and construction management • Safety, quality assurance and control • Programme and cost control	• Cleanroom classification testing • System performance verification • As-built and handover documents	• Tool move-in and placement • Utilities tied in to each tool • Qualified and released to production	• Planned preventive maintenance • Breakdown response • Optimisation and retrofits
Carried by						
EPC	■	■	■	■	■	■
PCU & TTI	■	■	■	■	■	when asked
EFM (Energy Facility Management)	when asked	when asked	when asked	■		■

Every discipline
Civil, structural and architectural, MEP (Mechanical, Electrical and Plumbing), and process utilities, in house.

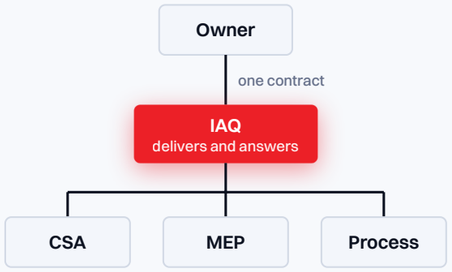
Any combination
One service, several, or all six. The model that governs it stays the same.

Two delivery models
EPCC, IAQ delivers the whole project. EPCM, IAQ manages it for the owner.

Short forms: BIM (Building Information Modelling) · EPC (Engineering, Procurement and Construction) · PCU & TTI (Process Critical Utilities & Total Tool Installation) · EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management)

Four ways to contract IAQ.

The model governs the work. The scope, one service or all six, sits inside it. In EPCC (Engineering, Procurement, Construction and Commissioning), IAQ does the work. In EPCM (Engineering, Procurement and Construction Management), IAQ manages the work for the owner.

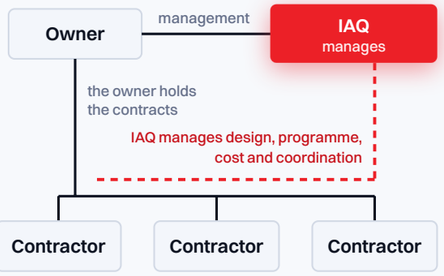


EPCC

IAQ delivers the complete project, commissioning included, and answers for how the system performs.

Contracts held by	IAQ
Answers for the result	IAQ
Typical term	12 to 24 months

Best when
You want one accountable party.

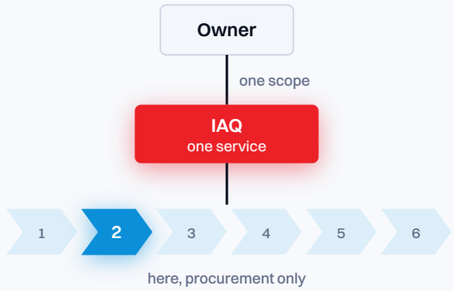


EPCM

The owner holds the construction contracts. IAQ manages design, contractors, programme and cost.

Contracts held by	The owner
Answers for the result	IAQ manages, the contractors deliver
Typical term	12 to 24 months

Best when
You want direct control of the contracts, often on the largest programmes.

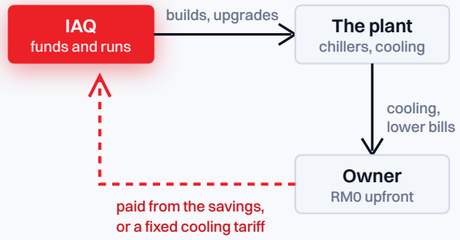


A single service

Any one of the six on its own: procurement only, design only, hookup only. The same model governs it.

Contracts held by	IAQ, for that scope
Answers for the result	IAQ, for that scope
Typical term	Set by the scope

Best when
You already hold the rest of the scope.



ESCO contract

EFM (Energy Facility Management) funds and runs the energy upgrade. Cooling as a Service, or Energy Performance Contracting.

Funded by	IAQ
Answers for the result	IAQ, for the savings
Typical term	5 to 20 years

Best when
You want lower energy cost with no capital spent.

Short forms: CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing) · ESCO (Energy Service Company)

Three disciplines. One contractor.

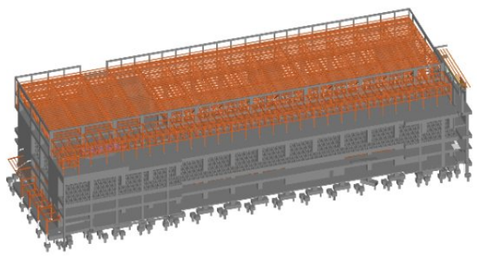
The market classifies a contractor by the work it delivers. A specialist carries one discipline. IAQ carries all three. The drawings are IAQ's own Revit model.

CSA

Civil, Structural and Architectural

The building itself.

- Piling and foundations
- Structural frame and roof
- Cleanroom envelope: walls, ceiling grid, raised floor
- Architectural finishes

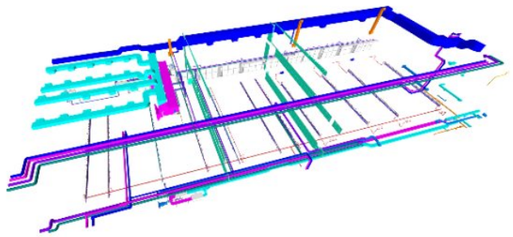


MEP

Mechanical, Electrical and Plumbing

What makes the building run.

- HVAC, ACMV and fan filter units
- Process cooling water (PCW)
- Chiller plant and district cooling
- Fire protection
- Electrical, high and low voltage
- Plumbing and drainage

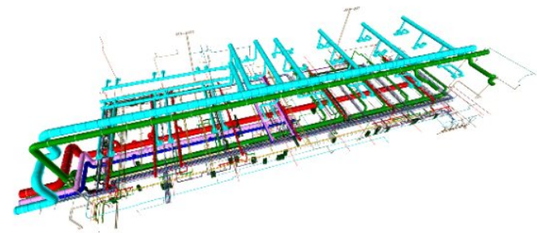


Process utilities

The utilities a production tool consumes

What the tool runs on.

- Clean dry air (CDA)
- Specialty gases and chemical delivery
- Ultrapure water (UPW)
- Process vacuum (PV) and exhaust
- Waste treatment



IAQ	Inclusive: civil to process, under one roof				
A civil contractor	CSA (Civil, Structural and Architectural) only	**An M&E contractor**	MEP (Mechanical, Electrical and Plumbing) only	**A process specialist**	Process only

Short forms: HVAC (heating, ventilation and air conditioning) · ACMV (air conditioning and mechanical ventilation) · M&E (mechanical and electrical)

EPC: one contract, from first drawing to final handover.

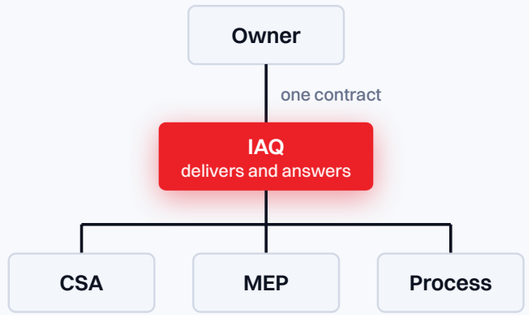
Engineering, Procurement and Construction. IAQ designs the system, procures it, builds it and proves it operates as intended.

1
point of contact for the whole facility

12 to 24
months, a typical EPC programme

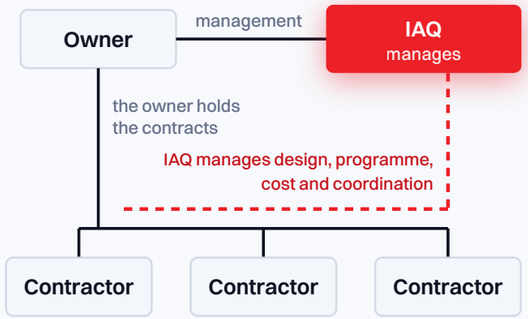
6
services, from design to tools hookup

3
disciplines in house: CSA, MEP, process



EPCC (Engineering, Procurement, Construction and Commissioning)

Turnkey. IAQ delivers the complete project and is accountable for how the system performs.



EPCM (Engineering, Procurement and Construction Management)

Managed for the owner, who keeps the construction contracts and direct control.

At a glance	EPCC	EPCM
Who holds the construction contracts	IAQ	The owner
Who answers for the system performing	IAQ	IAQ manages, the owner's contractors deliver
Commissioning and start-up	Included	Coordinated by IAQ
Typical programme	12 to 24 months	12 to 24 months

Short forms: CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing)

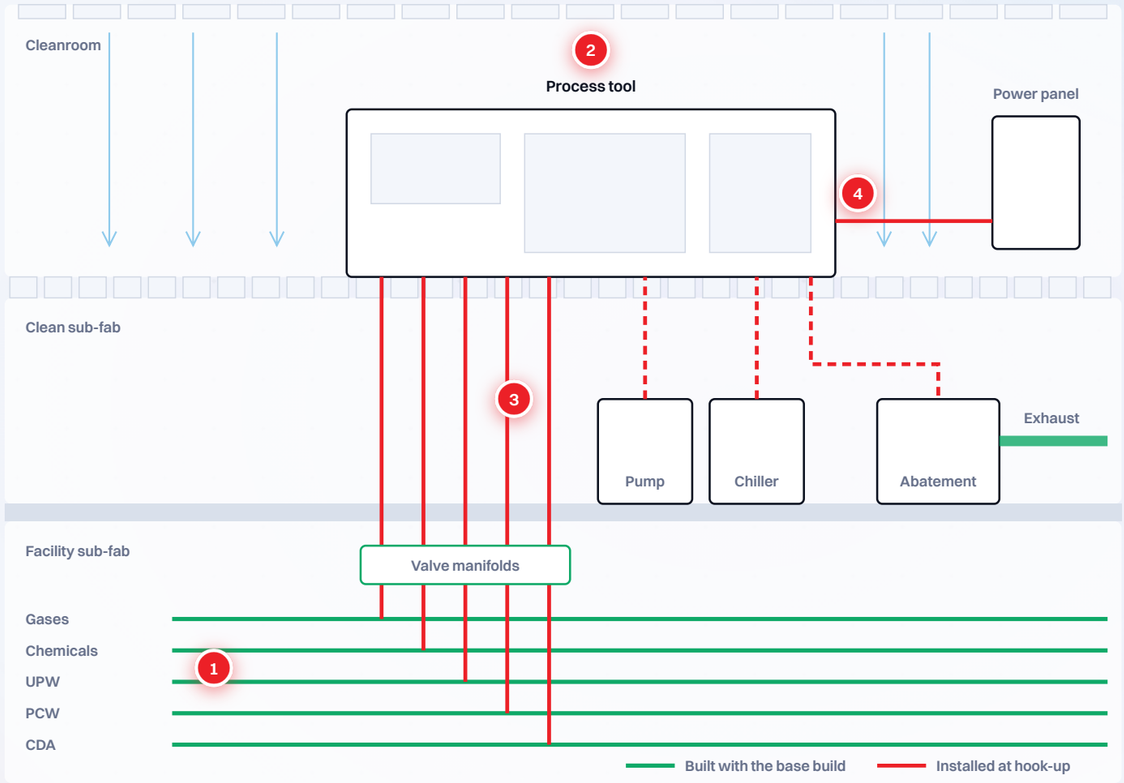
From a delivered tool to a producing tool.

Two halves, one unit, for semiconductor fabs only. Process Critical Utilities builds the utilities with the base build. Total Tool Installation then connects each tool to them.

Process Critical Utilities

IAQ's five solution groups

- **Water**
  UPW (ultrapure water) plant and distribution, process drains, waste water treatment, reclaim
- **Chemical and slurry**
  Bulk chemical, blending, dispense, storage tanks
- **Gas**
  Bulk and specialty gas, abatement, bunkers, liquid dispense
- **Process exhaust**
  General, acid, caustic, solvent and calamity
- **Space management**
  The sub-fab planned in 3D BIM (Building Information Modelling), off-site modules



Total Tool Installation

IAQ's four phases, marked on the section

1. **Facilitization**
   Before the tool arrives: drops, power, structure and room built to the maker's requirements.
2. **Rig-in and move-in**
   Uncrated in an airlock, moved on an agreed route, set, levelled and isolated.
3. **Hook-up**
   Each utility tapped from its valve manifold and connected into the tool.
4. **Commissioning**
   Leak tested, purged, balanced, trip tested, then handed to operations.

How it is bought

On its own, even in a fab IAQ did not build, or inside an EPCC (Engineering, Procurement, Construction and Commissioning) **or EPCM** (Engineering, Procurement and Construction Management) **contract.**

Short forms: PCW (process cooling water) · CDA (clean dry air)

Profile: extension page 3.

EFM: lower energy bills, funded by IAQ.

Energy Facility Management, a registered ESCO (Energy Service Company) under Suruhanjaya Tenaga. IAQ funds, builds and runs the upgrade.

Two ways to pay for it

10 to 20 years

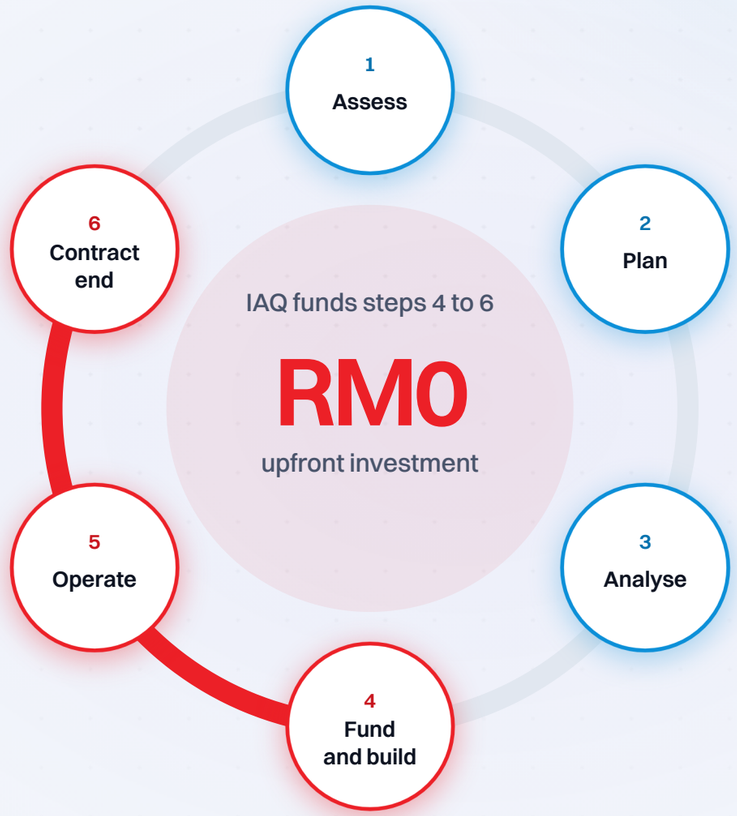
Cooling as a Service

IAQ finances, builds or rehabilitates, and operates the chiller plant. A fixed tariff for the cooling used, under a service level agreement. The plant is handed over at the end.

5 to 10 years

Energy Performance Contracting

IAQ funds and delivers the upgrades. Every payment comes from the savings actually measured. After the contract, the owner keeps all of them.



The four problems it solves

High electricity bills

Inefficient cooling, driving cost up year after year.

Rising carbon emissions

Compliance risk against regulation and targets.

Ageing infrastructure

Chiller plant past its best, and a capital bill to replace it.

Operational disruption

Downtime from old systems and poor maintenance.

Proof, by market and by the work delivered.

The 18 published references of 250+ delivered projects. Each light is one project; client names are withheld.

Market	Cleanroom	Dry Room	EPCC (Engineering, Procurement, Construction and Commissioning)	EPCM (Engineering, Procurement and Construction Management)	HVAC (heating, ventilation and air conditioning) / M&E (mechanical and electrical)	District Cooling
Semiconductor	●●●		●	●		
Data Centre					●	
EV Battery		●●			●	
Photovoltaics					●	
Bio LifeScience	●●●					
Food & Beverage	●●					
District Cooling & Heating						●●●

By region

Malaysia	14
Singapore	1
China	1
Europe	2

The full record is 250+ projects. This slide grows with the approved project list: every project added to the registry lights here on its own.

Watch it, by business unit

Nine short tutorials. Each one explains a term used on this page.

EPC (Engineering, Procurement and Construction)

PCU & TTI (Process Critical Utilities & Total Tool Installation)

EFM (Energy Facility Management)

1. **EPC and EPCM, explained** (Project Business, 6:17): youtube.com/watch?v=QdgS9MBDZSA
2. **What the ISO cleanroom classes mean** (Sure-BioChem Laboratories, 3:20): youtube.com/watch?v=IENbOfC4df8
3. **Inside the Morrow Batteries dry room** (Munters, 3:17): youtube.com/watch?v=dGd7hXcx_7w
4. **How TSMC builds a 19 billion dollar fab** (Asianometry, 12:50): youtube.com/watch?v=785Uzi1mGAA
5. **Inside an Intel sub fab** (Intel Newsroom, 3:02): youtube.com/watch?v=8jpRADL3BGM
6. **Ultrapure water for chip making** (Asianometry, 12:50): youtube.com/watch?v=C3RzODSR3gk
7. **District cooling in two minutes** (International District Energy Association, 1:46): youtube.com/watch?v=tEh16-NusuQ
8. **Cooling as a Service** (BASE Foundation, 2:31): youtube.com/watch?v=o27l1xsqiBw
9. **Energy Performance Contracting** (Codema, Dublin's energy agency, 6:33): youtube.com/watch?v=kWF0ywtEvhY

Six buyer requests, each read as a path through the layers

"A new fab, one party accountable for the result."

Unit	**EPC**
Model	EPCC
Scope	Design · Procure · Construct · Commission
Work	CSA · MEP · Process utilities
System	The whole facility

"We hold the contracts. Manage the programme for us."

Unit	**EPC**
Model	EPCM
Scope	Design · Procure · Construct · Commission
Work	CSA · MEP · Process utilities
System	The whole facility, through the owner's contractors

"Buy the process cooling water package for us, that one package."

Unit	**EPC**
Model	EPC, one service
Scope	Procure
Work	MEP
System	Process cooling water (PCW)

"Design, procure and build the PCW loop."

Unit	**EPC**
Model	EPC, three services
Scope	Design · Procure · Construct
Work	MEP
System	Process cooling water (PCW)

“Forty tools land in March. The fab is live.”

Unit	**PCU & TTI**
Model	Standalone, or under EPC
Scope	Design · Procure · Construct · Commission · Tool Installation
Work	MEP · Process utilities
System	Gases, chemicals, UPW (ultrapure water), exhaust and power, to each tool

“Cut the cooling bill, with IAQ funding the work.”

Unit	**EFM**
Model	Cooling as a Service
Scope	Maintain · Commission
Work	MEP
System	Chiller plant, run by IAQ under a tariff

3 On the website. How the site shows it, page by page.

The same order everywhere: business unit first, then how it is bought, what is in it, who does it, the proof, and one action. What is live now and what is proposed.

The order a visitor reads in

Every page reads in this order, and ends on one action.

1 Business unit
Which of the three you are buying from.
EPC · PCU & TTI · EFM

2 Model
How it is governed: EPCC, EPCM, standalone, ESCO.
EPCC · EPCM · Standalone · ESCO

3 Service
The scope in the contract, one or all six.
1 2 3 4 5 6

4 Work
The disciplines and the systems inside them.
CSA · MEP · Process · PCW · CDA · UPW

5 Proof
Projects delivered this way.
7 markets

6 One action
Contact. Once, at the end.
Contact

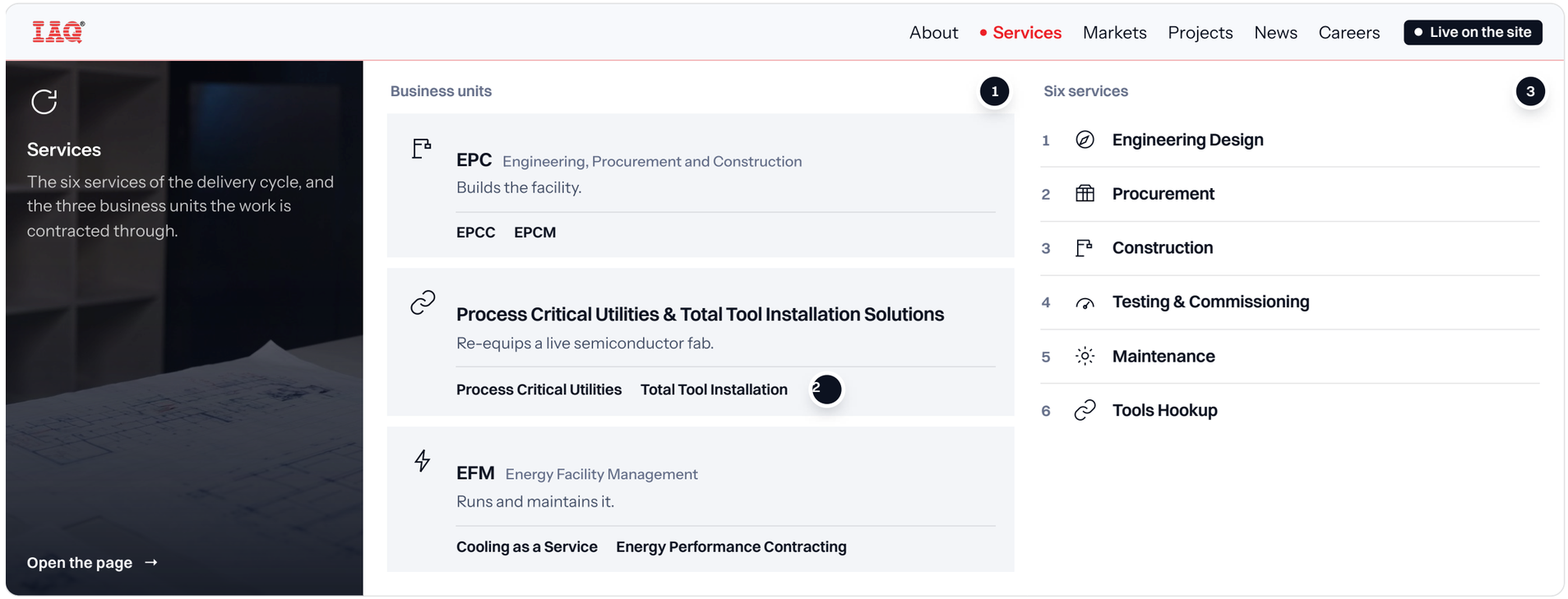
One ask from the review, read in order: “Buy the process cooling water package for us, that one package”

EPC → EPC, one service → 2 Procurement → MEP PCW → PCW projects, by market → Contact, once

Business unit · Delivery model · Service · Work · System · Market

The Services menu, live now

Three columns: the picture, the three units as cards, the six services as one list. Point at a unit to see what it carries.



1 Three columns: the picture, the units, the services.

The three business units are three cards, because a buyer chooses the unit first.

2 Every section inside a unit is a link.

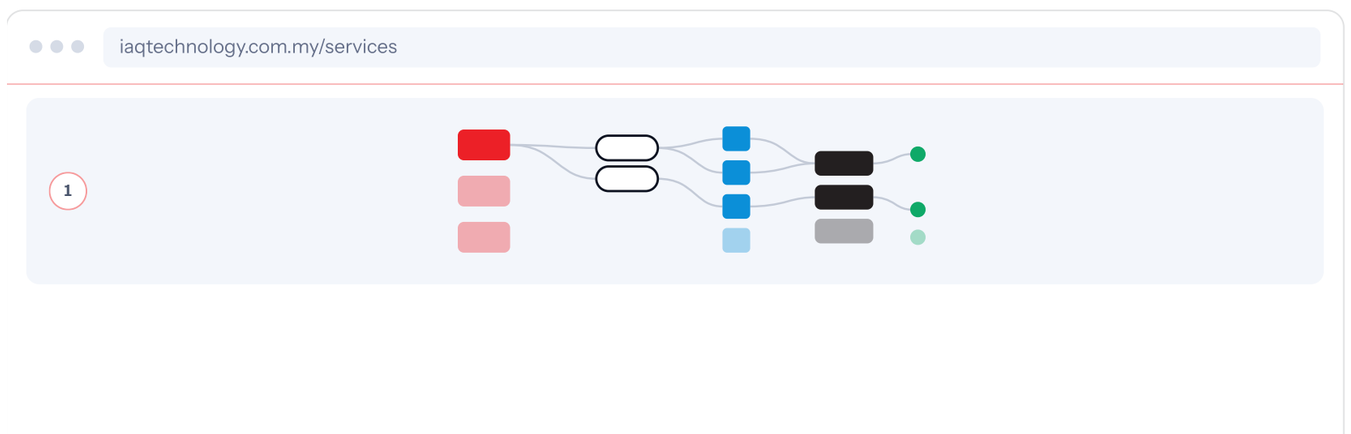
Its own pages, and its contract models, which open "Two ways to buy it" on the unit's page.

3 The six services, one numbered list.

Point at a unit and the list lights what it carries, and marks what it carries only when asked.

The Services page, top to bottom

Seven bands in the reading order, as the page stands on 26 September 2026. Point at a band to see its job, click it to see its slide.

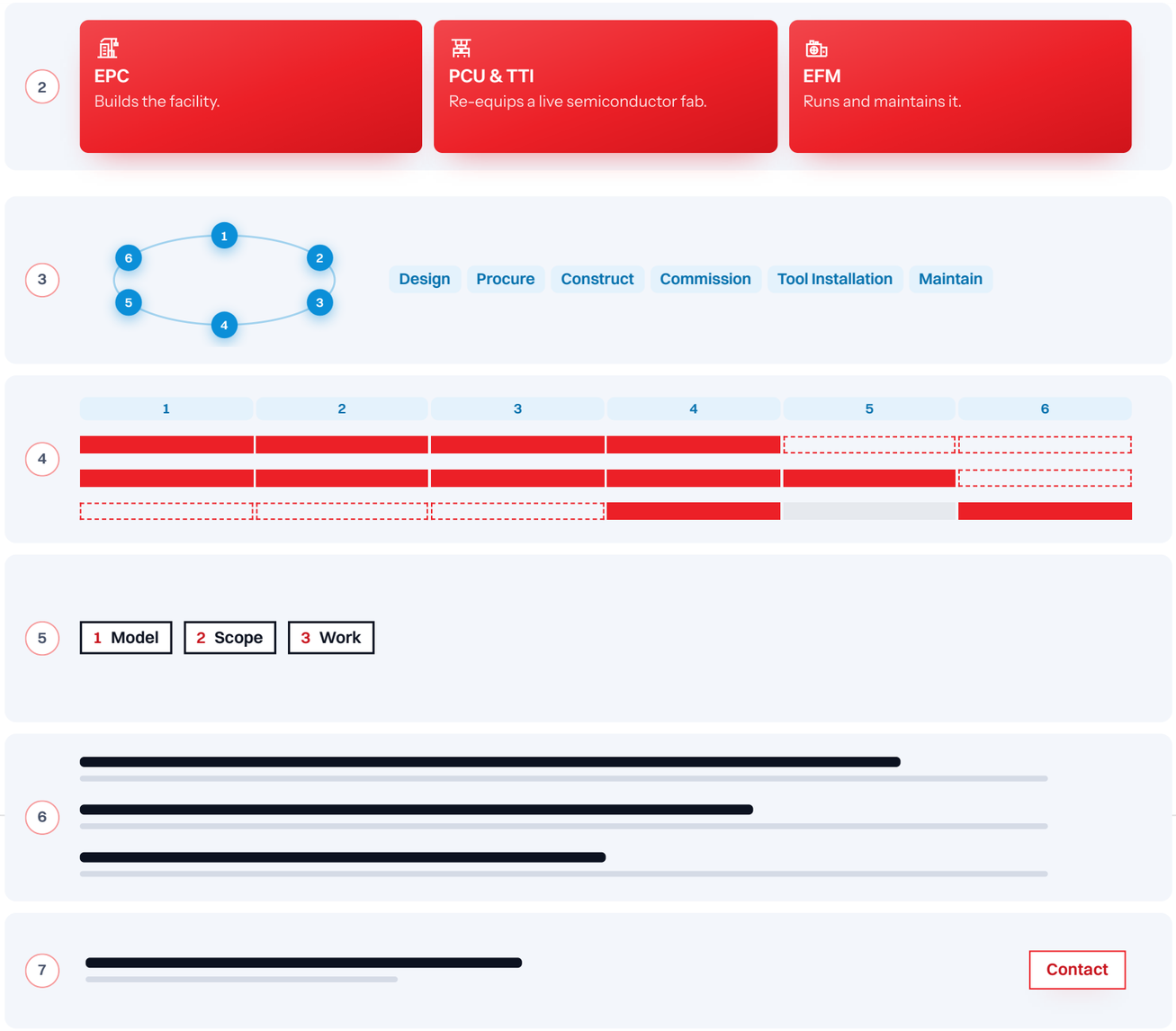


● Live built today ○ Proposed to build

The banner: the IAQ facility map ● Live

IAQ's own Revit model on the left (drag to turn, zoom, a tour you play), and on the right the map of 6 services, 3 business units, 3 kinds of work and 15 systems. Pick anything and the model builds to that moment and lights the layers it touches; the reading of the relation sits under the model, at one fixed height.

Business unit · Model · Service · Work



The three business units, one card each ● Live

White cards with the unit's photograph, its tag, its mark beside the name and its one line; then when to call it and how it is bought. Read more opens the three in detail as one bar across the cards. Each card opens its page.

Business unit

Six services, one accountable team ● Live

The horizontal serpentine: 1 to 6 left to right, the U-turn, the red return from Hookup into the next Design. It steps on its own until touched. Each stage opens its page.

Service

Six services, three units, four kinds of work, in one view ● Live

The one-view figure: the six services across, the three units as bars over the services they carry (solid always, dashed when asked), the kinds of work with who does each.

Model Service Work

What to settle before you ask for a quote ● Live

Three questions (model, scope, work) and six typical requests read as a path through the layers.

Model

Questions clients ask, answered ● Live

The questions in five groups (choosing a model, time and cost, a live fab, after handover, working with IAQ), each answer one paragraph with the page that says more. Codex part 1 holds the same set.

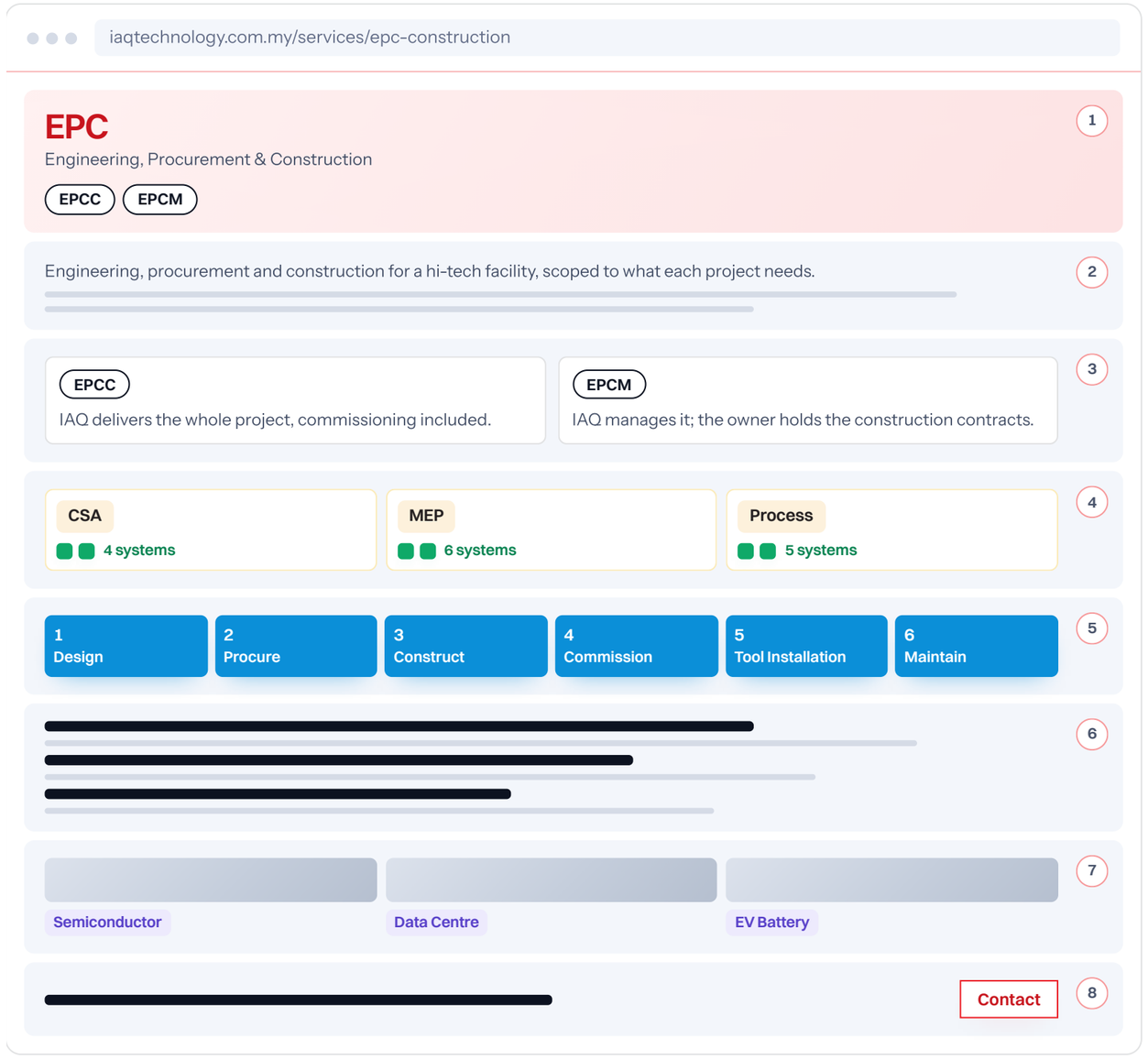
One action: contact ● Live

The closing band. The only call to action on the page.

One action

A business unit page, part by part

Eight parts, the same on all three unit pages, so they compare line for line; EFM carries its seven services where the models stand. EPC shown.



1. **Name and models**
   EPC, its facts and its photograph. Bought as EPCC or EPCM.
2. **What it is**
   In plain terms, the unit's own words; on EFM, the problems it solves under their own heading.
3. **The models**
   One card each: who holds the contracts, who answers for the result, the term. On EFM, the seven energy management services stand here instead.
4. **The work**
   CSA, MEP, process, and the systems under each; on EFM, district cooling in brief first.
5. **The cycle**
   The six stages beside the unit's drawing; each stage names what IAQ carries there.
6. **Why owners choose it**
   Six claims, one line each; a click opens the sentence behind it and never closes it.
7. **Proof**
   Projects tagged with this model, by scope and location.
8. **One action**
   Contact, once.

Where every picture goes

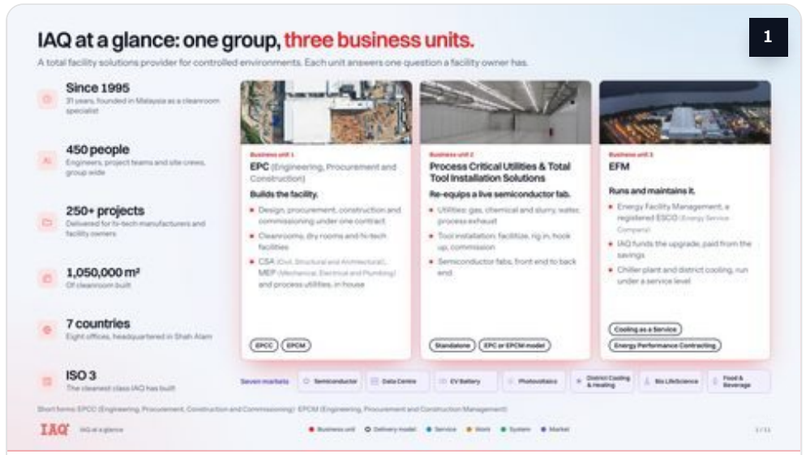
The twelve slides, and where each one is placed. Click a slide to open it full size.



Cover · One fab, every layer

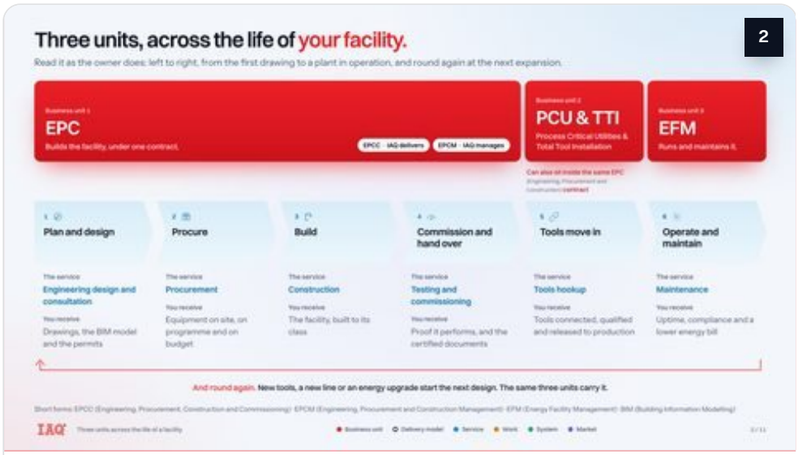
- **On the site** The Services page hero.
- **In the profile** Opens the business section.

Booth screen, beside the 3D model.



IAQ at a glance

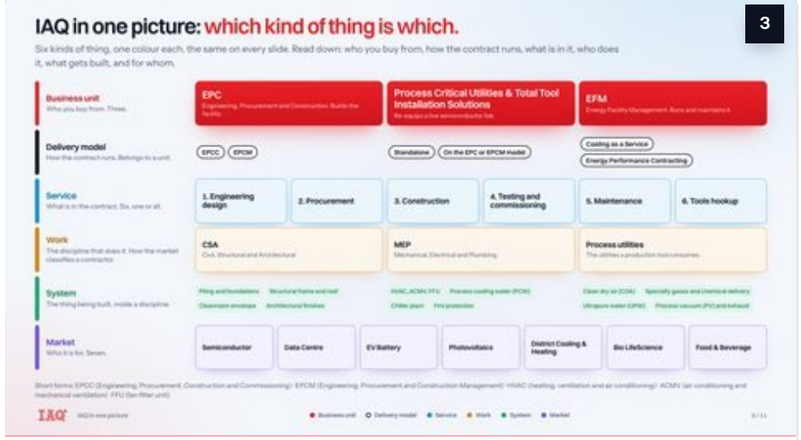
- **On the site** About.
- **In the profile** The opening spread.



Three units across the life of a facility

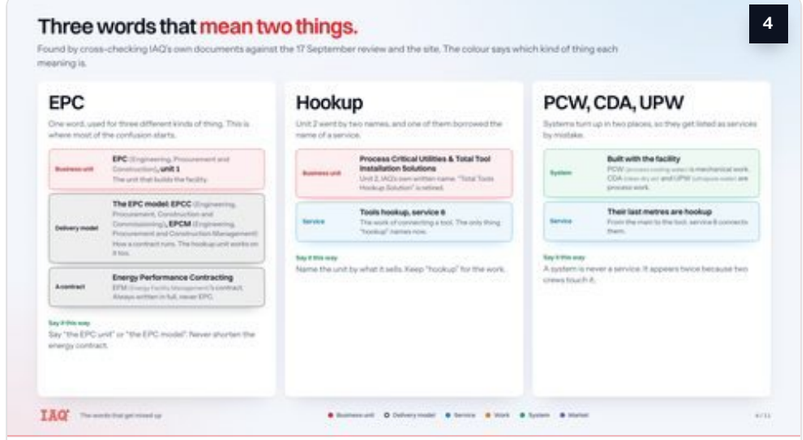
- **On the site** Services, first screen.
- **In the profile** The page on the three business units.

Booth screen.



IAQ in one picture

- **On the site** Services, beside the relationship map.
- **In the profile** Before the scope of services.



The words that get mixed up

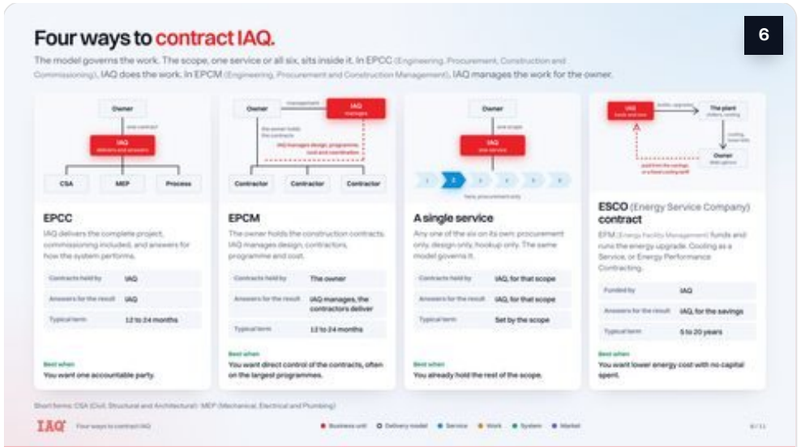
- **On the site** Not on the site.
- **In the profile** Not in the profile.

Internal first: the team, the profile writers, the booth crew. Then the glossary's place on the site.



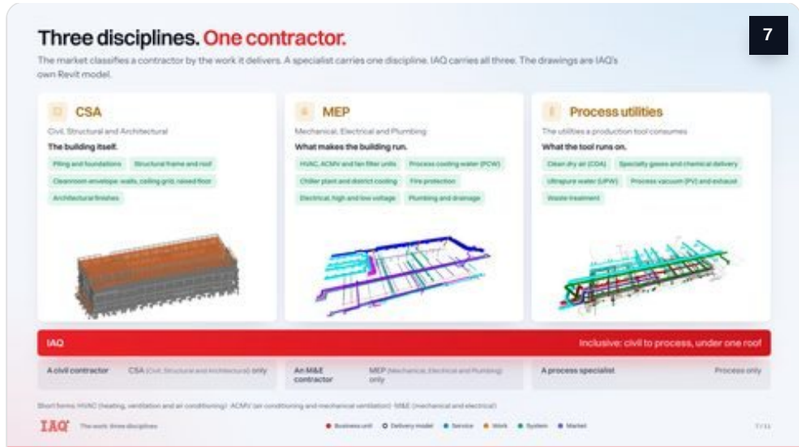
Six services

- **On the site** Services, under the units.
- **In the profile** Scope of services.



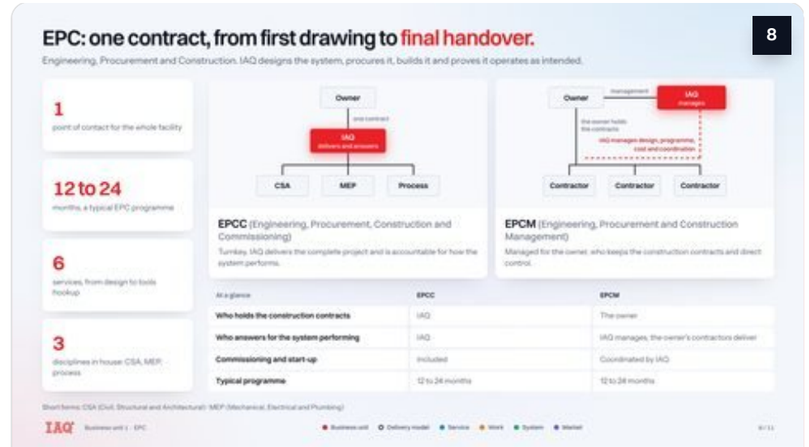
Four ways to contract IAQ

- **On the site** — The EPC page.
- **In the profile** — After scope of services.



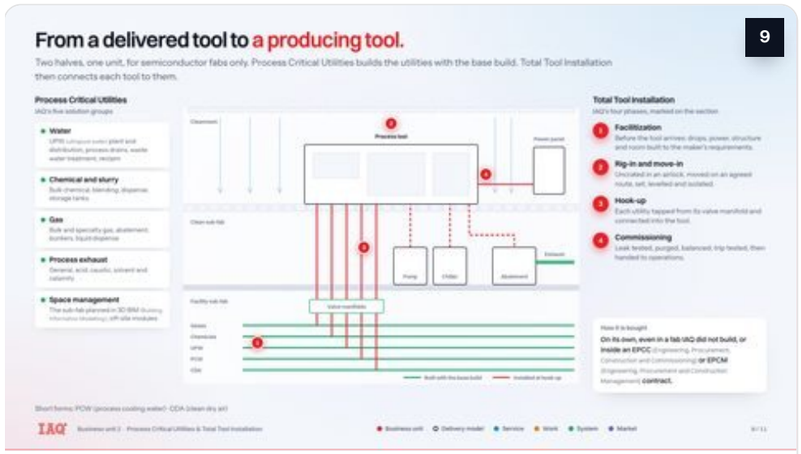
The work: three disciplines

- **On the site** — Services, the glossary's place.
- **In the profile** — Scope of work.



Business unit 1 · EPC

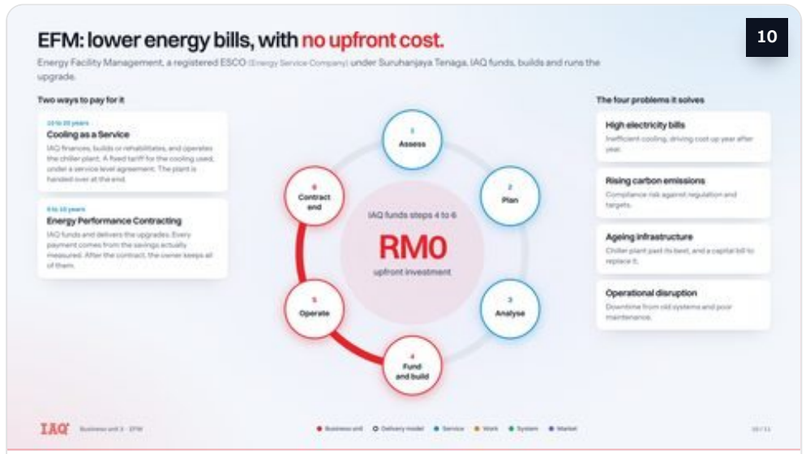
- **On the site** — Not on the site.
- **In the profile** — Extension page 1, after the page on the three business units.



Business unit 2 · Process Critical Utilities & Total Tool Installation

- **On the site** — Not on the site.
- **In the profile** — Extension page 2.

Booth screen beside the 3D model.



Business unit 3 · EFM

- **On the site** — Not on the site.
- **In the profile** — Extension page 3.



Proof, by market and by work delivered

- **On the site** — Projects.
- **In the profile** — Opens the project references.

Interactive moments to build

Four interactive pieces that show how the parts connect. Two are live on the site, two are proposed.



The fab build-up

Home page, in 3D. Press play and the fab assembles layer by layer, each layer named by its work.

Proposed Asked for on 17 Sep



The relationship map

Services page. Pick a unit, and its models, services, work and systems light up.

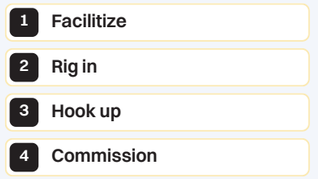
Live



The scope lights

Each unit page. The six services, the unit's own lit, the ones carried on request ringed.

Live



The hook-up section

Total Tool Installation page. IAQ's four phases, from facilitization to commissioning.

Live

One name each, one goal per page

The names

~~EPC & Construction~~ → **EPC, Engineering, Procurement and Construction**

EPC already includes construction. Nabilah, 17 Sep.

~~Total Tools Hookup Solution~~ → **Process Critical Utilities & Total Tool Installation Solutions**

IAQ's own name in the Discovery answers, the website copy sheet, the questionnaire and Nabilah's 18 Sep message. "Hookup" stays the name of service 6, so one word no longer names two things.

~~Energy Management~~ → **EFM, Energy Facility Management**

Nabilah's 18 Sep name for unit 3. The Discovery answers say Energy Management Solutions; the questionnaire confirms it covers what EFM provides.

~~EPC, for the energy contract~~ → **Energy Performance Contracting, always spelled out**

EPC means Engineering, Procurement and Construction everywhere else.

The goals

Home	Understand what IAQ is, then choose a business unit.
Services	Choose the business unit.
A unit page	Contact, knowing the model, scope and work.
Markets	Find your own sector, then its proof.
Projects	Filter the record to work like yours.
About	Trust: the record, the certifications, the people.
News	Read the latest.
Careers	Apply.

In the company profile

IAQ, 18 September: the three unit pages go into the profile straight after the page on the three business units. Every picture above has its place.

C 1	**Opening** The cover, then who IAQ is in one view.
2	**The page on the three business units** The three units along the life of the owner's facility.
8 9 10	**Directly after it, the extension pages IAQ asked for on 18 Sep** EPC (Engineering, Procurement and Construction), then Process Critical Utilities & Total Tool Installation, then Energy Facility Management.
3 5 6 7	**Scope of services** The one picture, the six services, the four ways to contract, the three disciplines.
11	**Opening the project references** Proof by market and by work delivered.

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · ESCO (Energy Service Company) · CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing) · CDA (clean dry air) · UPW (ultrapure water)

IAQ's own documents, checked line by line.

The Discovery answers, the three business unit questionnaires, the website copy sheet, the Utility Solutions deck, the moodboard, the annotated feedback deck, the 17 September review and Nabilah's 18 September message, read against the site. Where they disagree, IAQ's written word wins.

5 Fixed on the site | **6** Already agreed | **3** IAQ's papers disagree

Fixed on the site

The name of business unit 2

IAQ's documents

Discovery answers A2.1 Process Critical Utilities & Total Tool Installation Solutions

Website copy sheet the same

Nabilah, 18 Sep message Process Critical Utilities and Total Tool Installation Solutions

17 Sep review flags two names in use

The site said

Total Tools Hookup Solution, from the 7 Aug meeting

Now

IAQ's written name everywhere. Hookup stays the name of service 6 only.

Fixed on the site

The name of business unit 3

IAQ's documents

Discovery answers A2.1 Energy Management Solutions

EFM questionnaire Energy Management covers what EFM provides

Nabilah, 18 Sep message Energy Facility Management

The site said

Energy Management, then EFM

Now

EFM, Energy Facility Management, the latest name IAQ used.

Fixed on the site

Unit 2, the sector

IAQ's documents

Unit 2 questionnaire Semiconductor exclusively

Utility Solutions deck Front end, advanced packaging and back end fabs

The site said

Semiconductor and pharma, on the Total Tool Installation page

Now

Semiconductor only, on the page, the utilities page and the slides.

Fixed on the site

Unit 2, the hook-up sequence

IAQ's documents

Unit 2 questionnaire Four phases: facilitization, rig-in, hook-up, commissioning

Utility Solutions deck A six-phase fab hook-up timeline, base build and tool vendor side by side

The site said

Six generic steps written from the plan brief

Now

IAQ's four phases, with the unit head's own detail under each.

Already agreed

Unit 2, what the utilities are

IAQ's documents

Unit 2 questionnaire Chemical, gases, ultrapure water, deionised water, waste treatment

Utility Solutions deck Water, chemical and slurry, gas, process exhaust, plus space management in BIM (Building Information Modelling)

The site said

Gases, chemicals, ultrapure water, exhaust

Now

The slide groups them as IAQ's deck does.

Already agreed

Unit 2, how it is bought

IAQ's documents

Unit 2 questionnaire Standalone, but using the EPC (Engineering, Procurement and Construction) or EPCM (Engineering, Procurement and Construction Management) model

Utility Solutions deck A specialist EPCM partner

The site said

Standalone, or on the EPC or EPCM model

Now

Unchanged.

IAQ's papers disagree

How many business units

IAQ's documents

Discovery answers A2.1 Three business models

Website copy sheet Headline "Four Ways We Deliver Certainty", body "three core business models", notes "4 clickable model cards"

The site said

Three

Now

Three. The copy sheet's "four" reads as a leftover.

Already agreed

What the six are called

IAQ's documents

Discovery answers A1.3 Not core services: the Total Solutions, six, Tools Hookup added

17 Sep review The six services are the scope

The site said

Six services

Now

Six services, subtitled IAQ's total solution.

Already agreed

EPC and EPCM

IAQ's documents

EPC questionnaire EPC is one-stop; EPCM manages the project for the owner

17 Sep review EPC does the work, EPCM manages the work; high-end MNCs engage EPCM

The site said

The same

Now

Unchanged.

IAQ's papers disagree

The founding year

IAQ's documents

Discovery answers A2.3 1994

Website copy sheet Since 1994, 32 years

17 Sep review Brand book corrected 1994 to 1995

The site said

1995, on Nabilah's instruction of 10 Sep

Now

1995 kept. IAQ's own papers still say 1994.

IAQ's papers disagree

How many countries

IAQ's documents

Discovery answers A2.3 Malaysia, Singapore, Sweden, India, Germany

Website copy sheet 8 countries

The site said

7 countries, on Nabilah's instruction of 10 Sep

Now

7 kept, until IAQ confirms the list.

Already agreed

Markets, names and order

IAQ's documents

Discovery answers A1.4 Semiconductor, Data Centre, EV Battery, Photovoltaics, District Cooling & Heating, Bio LifeScience, Food & Beverages

Website copy sheet Pharmaceuticals, six cards

The site said

DV3's seven, in DV3's order

Now

Unchanged. The copy sheet is the older list.

Fixed on the site

Colour

IAQ's documents

Feedback deck This page is just too white

Moodboard Keep the corporate colour, add one or two bold colours that complement it; hi-tech tone

The site said

The first infographic set, white and red only

Now

The set carries one bold colour per kind of thing, on a light ground, with soft glows.

Already agreed

Subsidiary names

IAQ's documents

Discovery answers A2.1 We do not need to disclose subsidiary names, only the business model

Utility Solutions deck Titled IAQ Utility Solutions

The site said

No entity names on public pages

Now

Unchanged. The deck title stays internal.

Background

The competitor benchmark IAQ sent, the reading rules, every term placed by layer, and what is applied or next.

The benchmark: what Sunny expects.

On 18 September IAQ forwarded a competitor's company introduction deck with one line: this is the expectation, infographics that help clients understand the business. Read slide by slide, it does six things our first Codex did not.

1 One idea per slide
The headline states it. Nothing else competes.

2 A drawing at the centre
Text sits round the drawing. No grid of text tiles.

3 Models inside the service picture
EPC (Engineering, Procurement and Construction) and EPCM (Engineering, Procurement and Construction Management) are named where they apply, not on a page of their own.

4 Proof coded the way a buyer classifies
By delivery type and discipline, with counts.

5 Every acronym spelled out
In a legend on the slide that uses it.

6 Numbers with an icon and a unit
Never a bare figure.

Their deck, slide by slide, and the IAQ slide that answers it

Their slide	The device	IAQ's answer	Where we stand
Group overview	Six facts with icons on the left; the three business units as columns on the right, each with its sectors.	Slide 1	Their columns carry share of sales. IAQ keeps revenue unpublished, so ours carry what each unit does, under IAQ's own photographs.
Global presence and performance	A world map, a workforce ring by function, three KPI tiles, two rings for sales by unit and by region.	Not in the set	Not in this set. Needs numbers IAQ has not released: headcount by function, the safety rate, order intake. The site's Global Presence page carries the map.
Industries and structure	An organisation tree: the group, its EPC arm with three business units and their sectors, a second arm for equipment and services, a heritage photograph.	Slide 1 Slide 2	Ours reads the three units along the life of the owner's facility, which also says when each one is called.

Their slide	The device	IAQ's answer	Where we stand
Full spectrum of services	Five chevrons in two phases, the scope as bullets under each, the contract models named inside the stage they apply to, three claims at the foot.	Slide 5 Slide 6	This is the slide that answers Nabilah's 17 Sep point. Ours adds the business unit that carries each service, drawn under the bar.
Regional project record	A map with every project coded by delivery type, a count per type, and a legend that spells out each acronym.	Slide 11	Ours is a matrix, market by work delivered, computed from the registry. It becomes as strong as theirs when the approved project list arrives.
Positioning	One sentence, then three linked circles: end to end, predictable, proven.	Not in the set	The site hero and the closing band already say this. One slide if the profile wants it.
Value chain coverage	The client's own process from front end to back end, with one proof project under each part.	Slide 2	Ours is the life of the facility rather than the life of the wafer, because IAQ sells to the facility owner.
Integrated installation	Capabilities and delivery value as two short lists beside a site photograph.	Slide 9	The unit pages on the site already do this with IAQ's photographs.
Tool installation	A cutaway of a three-level fab, numbered callouts on the drawing tied to a numbered list beside it.	Slide 9	Ours is a section, with the base build in grey and what the hookup unit installs in red.
Modular solutions	Four numbered panels, a picture and three bullets each.	Not in the set	The unit pages' step rails cover this; no slide needed yet.
Facility services lifecycle	A ring of services round a photograph, five numbers on the left, five pillars on the right.	Slide 10	Ours rings the six contract steps round the one number that matters to the buyer: no upfront investment.

The reading rules

EPC is a model, not a service.

It governs how a project is delivered. The six services are the scope inside it. "EPC and Construction" repeats itself: EPC already includes construction.

A service can be bought alone. The model still governs it.

Procurement only, design only, or design, procure and construct: each is a scope under the EPC model.

PCW and CDA are systems, not services.

Process cooling water sits in mechanical work. Clean dry air sits in process work. Neither belongs in a service list.

Hookup is work. The contract can be EPC or EPCM.

Tool hookup runs under either model, and it can be bought on its own.

EPC does the work. EPCM manages the work.

The difference is who holds the construction contracts. Large multinational clients engage EPCM on the biggest programmes.

Engineers read labels, not descriptions.

Procurement is "Procurement". Commissioning is "Commissioning". One line each, no paragraph repeating the word.

The terms, placed by layer

Models

EPC
Engineering, Procurement and Construction. The project delivery model, and the name of unit 1.

EPCC (Engineering, Procurement, Construction and Commissioning)
EPC plus Commissioning. IAQ delivers the whole project.

EPCM
EPC Management. IAQ manages the project; the owner holds the construction contracts.

ESCO
Energy Service Company, registered with Suruhanjaya Tenaga. The status EFM (Energy Facility Management) works under.

CaaS
Cooling as a Service. IAQ funds and runs the chiller plant for a fixed tariff.

Energy Performance Contracting
IAQ funds the upgrade and is paid from measured savings. Always spelled out on the site, never shortened to EPC.

Units

EPC
Unit 1. Builds the facility.

Process Critical Utilities & Total Tool Installation Solutions
Unit 2, PCU and TTI for short. Re-equips a live semiconductor fab. Its deck is titled IAQ Utility Solutions; entity names stay off public pages.

EFM
Unit 3. Energy Facility Management. Runs and maintains the facility.

Services

The six
Engineering design and consultation, Procurement, Construction, Testing and commissioning, Tool Installation, Maintenance.

T&C
Testing and Commissioning, service 4.

Tool Installation
Service 5, the work of connecting a tool. Never the name of a unit.

Work

CSA
Civil, Structural and Architectural.

MEP
Mechanical, Electrical and Plumbing. Also written M&E (mechanical and electrical).

Process
The utilities a production tool consumes: gases, chemicals, water, exhaust.

Systems

PCW

Process cooling water. Mechanical work.

CDA

Clean dry air. Process work.

UPW

Ultrapure water. Process work.

PV

Process vacuum. Process work.

HVAC, ACMV

Heating, ventilation and air conditioning; air conditioning and mechanical ventilation. Mechanical work.

FFU

Fan filter unit, the ceiling unit that makes a cleanroom clean.

BIM

Building Information Modelling, the coordinated 3D model every discipline is designed in.

Applied on the site

- The three business units lead the Services page; the six-service card set below them is gone, the cycle diagram carries the six with one line each. 17 Sep.
- The repeated “Start a project” band, the next-stage cards and the “where this goes next” strips are off every page. The closing band holds the one action. 18 Sep.
- Careers is one page: culture at the top, roles at the bottom, two entries in the menu. 18 Sep.
- The capability pack left the public Services page for the member portal. 18 Sep.
- The Services menu is three columns: the picture, the three units as cards (every section inside one a link, the contract models included), and the six services as one numbered list that lights what the unit under the pointer carries. No count or title block in any dropdown. 22 Sep.
- One name each, cross-checked against IAQ’s own documents: EPC; Process Critical Utilities & Total Tool Installation Solutions; EFM. 18 Sep.
- Unit 2 is semiconductor only and hook-up runs in IAQ’s own four phases, on the pages and the slides. 18 Sep.
- The member portal is its own area with tabs: Codex, Newsroom, Careers, Projects, Downloads. 18 Sep.
- Every unit page carries the scope and the work: the six services with the unit’s own lit, and the three disciplines with their systems. 18 Sep.
- The infographic set: a cover and eleven slides, one colour per kind of thing, on a light ground with soft glows, drawn to the benchmark IAQ sent on 18 Sep. Three of them are the unit extension pages for the company profile. 18 Sep.
- The closing band on every page is three sections: the ask with the contact lines beside it, the sitemap under its six main pages, and the sign-off. Careers uses the same band, pointed at HR. 22 Sep.

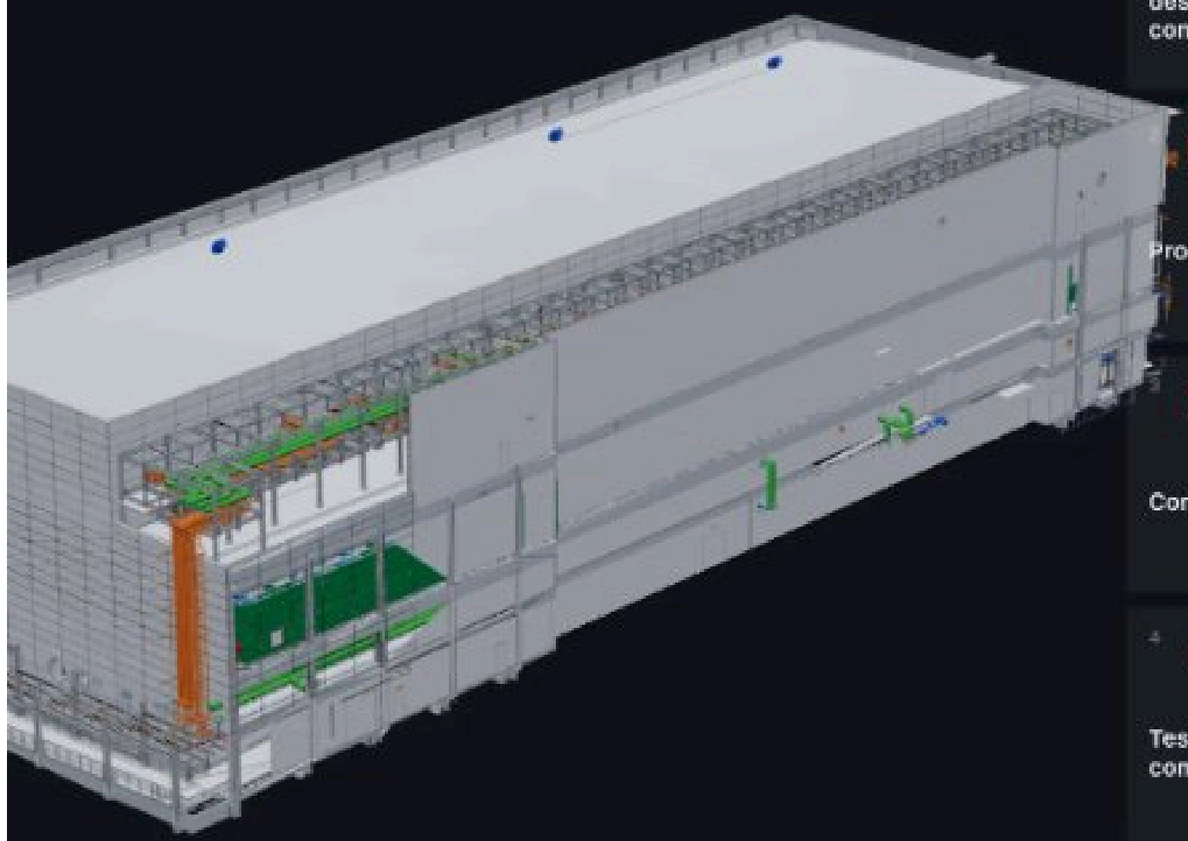
Next

- [ ] Place the three unit pages (slides 8, 9 and 10) in the company profile straight after the page on the three business units, and the rest in the order set out below. Shazwan.
- [ ] A responsive version of slides 2 and 5 for the public Services page.
- [ ] The cover and slides 2 and 9 on the booth screen beside the 3D model, for the 10 October backdrop and the 5 November screen content.
- [ ] The unit names in the company profile and the deck, to match the site.
- [ ] The 3D section on the home page shows the work by system type: civil and structural, architectural, process utilities, fire protection, air conditioning, electrical, cleanroom systems.
- [ ] The six service pages: keep them as depth reached from the cycle, or fold them into the Services page. IAQ to confirm.

Facility map, version 1. As it stood on 25 September 2026.

Four columns beside IAQ's Revit fab: service, business unit, work and system, with a reading panel under the model. Kept here before the simplified version 2. The code is kept in `src/_versions/facility-map-v1-2026-09-25`, with a note on how to put it back.

The IAQ facility map



+ − ▶ Play the tour

Choose a unit, a service, a kind of work or a system. The model builds to that moment and names the layers it touches; this panel reads the relation.

Service

1 Engineering design and consultation

Procurement

3 Construction

4 Testing and commissioning

5 Maintenance

6

Business unit

Unit 1

EPC
(Engineering, Procurement and Construction)
Builds the facility.

EPCC EPCM

Unit 2

Process Critical Utilities & Total Tool Installation Solutions
(PCU & TTI)
Re-equips a live semiconductor fab.

Standalone
On the EPC or EPCM model

Unit 3

EFM
(Energy Facility Management)
Runs and maintains it.

Cooling as a Service

Work

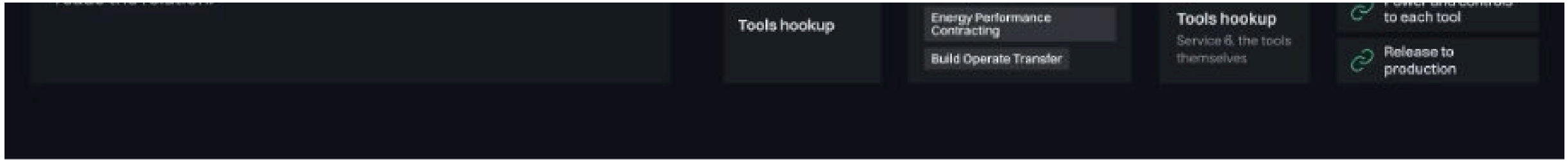
CSA
(Civil, Structural and Architectural)

MEP
(Mechanical, Electrical and Plumbing)

Process utilities
The utilities a production tool consumes

System

- Piling and foundations
- Structural frame and roof
- Cleanroom envelope
- Architectural finishes
- Air conditioning and ventilation (HVAC, ACMV), fan filter units
- Process cooling water (PCW)
- Chiller plant and district cooling
- Fire protection
- Electrical, high and low voltage
- Plumbing and drainage
- Clean dry air (CDA)
- Specialty gases and chemical delivery
- Ultrapure water (UPW)
- Process vacuum (PV) and exhaust
- Waste treatment
- Gas, chemical, water and exhaust connections to each tool
- Power and controls


Tools hookup
Energy Performance Contracting
Build Operate Transfer
Tools hookup
Service 6. the tools themselves
to each tool
Release to production

At rest

The IAQ facility map



+ - ▶ Play the tour

EPC (Engineering, Procurement and Construction) builds the facility. Bought as **EPCC** (Engineering, Procurement, Construction and Commissioning) **or EPCM** (Engineering, Procurement and Construction Management). It carries **engineering design and consultation, procurement, construction and testing and commissioning.** Its work is **CSA** (Civil, Structural and Architectural), **MEP** (Mechanical, Electrical and Plumbing) **and Process utilities.**

Service

1 **Engineering design and consultation**

2 **Procurement**

3 **Construction**

4 **Testing and commissioning**

5 **Maintenance**

6

Business unit

Unit 1

EPC
(Engineering, Procurement and Construction)
Builds the facility.

EPCC EPCM

Unit 2

Process Critical Utilities & Total Tool Installation Solutions
(PCU & TTI)
Re-equips a live semiconductor fab.

Standalone
On the EPC or EPCM model

Unit 3

EFM
(Energy Facility Management)
Runs and maintains it.

Cooling as a Service

Work

CSA
(Civil, Structural and Architectural)

MEP
(Mechanical, Electrical and Plumbing)

Process utilities
The utilities a production tool consumes

System

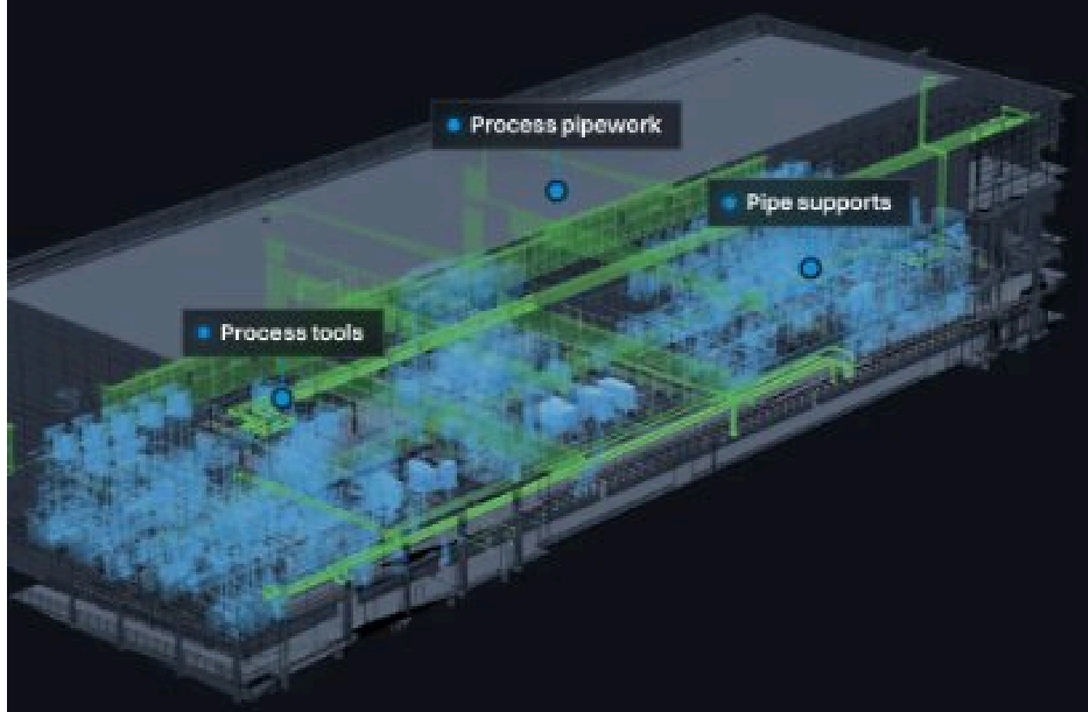
- Piling and foundations
- Structural frame and roof
- Cleanroom envelope
- Architectural finishes
- Air conditioning and ventilation (HVAC, ACMV), fan filter units
- Process cooling water (PCW)
- Chiller plant and district cooling
- Fire protection
- Electrical, high and low voltage
- Plumbing and drainage
- Clean dry air (CDA)
- Specialty gases and chemical delivery
- Ultrapure water (UPW)
- Process vacuum (PV) and exhaust
- Waste treatment

Gas, chemical, water and exhaust connections to each tool

Power and controls

EPC picked: design to commissioning

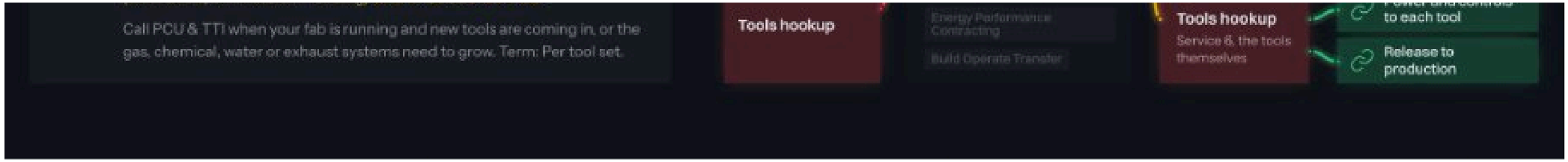
The IAQ facility map



+ - ▶ Play the tour

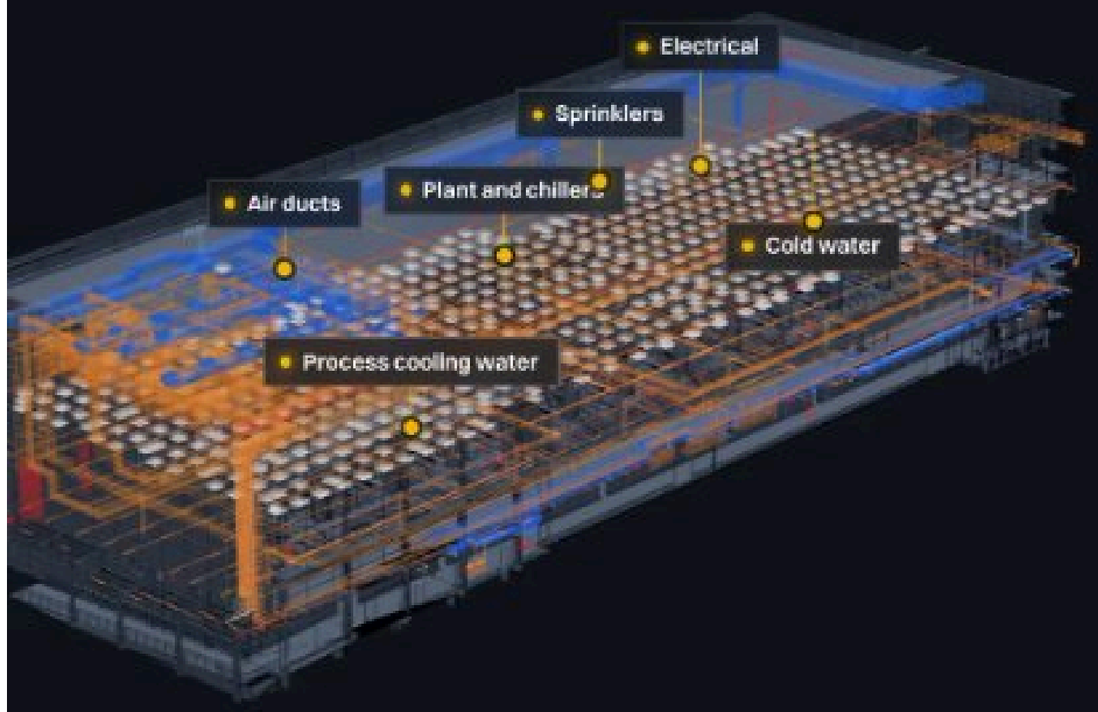
Process Critical Utilities & Total Tool Installation Solutions re-equips a live semiconductor fab. Bought as **a standalone job, or inside an EPC** (Engineering, Procurement and Construction) **or EPCM** (Engineering, Procurement and Construction Management) **contract**. It carries **engineering design and consultation, procurement, construction, testing and commissioning and tools hookup**, and **maintenance** when the client asks. Its work is **MEP** (Mechanical, Electrical and Plumbing) **and Process utilities**.

Service	Business unit	Work	System
1 Engineering design and consultation	Unit 1 — EPC (Engineering, Procurement and Construction) Builds the facility.	CSA (Civil, Structural and Architectural)	Piling and foundations
2 Procurement	EPCC EPCM		Structural frame and roof
3 Construction	Unit 2 — Process Critical Utilities & Total Tool Installation Solutions (PCU & TTI) Re-equips a live semiconductor fab. Standalone On the EPC or EPCM model	MEP (Mechanical, Electrical and Plumbing)	Cleanroom envelope
4 Testing and commissioning			Architectural finishes
5 Maintenance	Unit 3 — EFM (Energy Facility Management) Runs and maintains it.	Process utilities The utilities a production tool consumes	Air conditioning and ventilation (HVAC, ACMV), fan filter units
6	Cooling as a Service		Process cooling water (PCW)
			Chiller plant and district cooling
			Fire protection
			Electrical, high and low voltage
			Plumbing and drainage
			Clean dry air (CDA)
			Specialty gases and chemical delivery
			Ultrapure water (UPW)
			Process vacuum (PV) and exhaust
			Waste treatment
			Gas, chemical, water and exhaust connections to each tool
			Power and controls



PCU & TTI picked

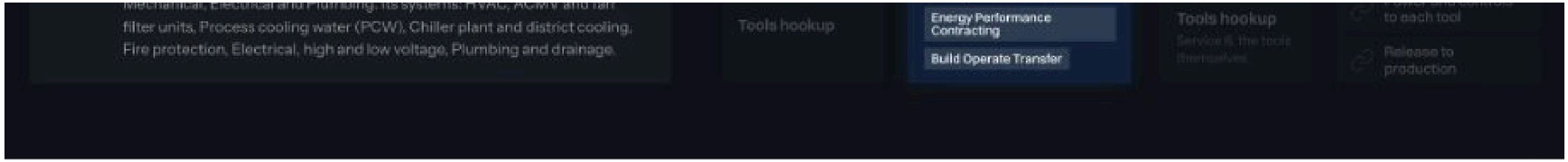
The IAQ facility map



+ - ↻ ▶ Play the tour

MEP (Mechanical, Electrical and Plumbing): what makes the building run. Delivered by **EPC** (Engineering, Procurement and Construction), **PCU & TTI** (Process Critical Utilities & Total Tool installation) and **EFM** (Energy Facility Management). Its systems include **HVAC** (heating, ventilation and air conditioning), **ACMV** (air conditioning and mechanical ventilation) and **fan filter units, process cooling water (PCW) and chiller plant and district cooling.**

Service	Business unit	Work	System
Engineering design and consultation	**Unit 1** EPC (Engineering, Procurement and Construction) Builds the facility. EPCC EPCM	CSA (Civil, Structural and Architectural)	Piling and foundations
Procurement			Structural frame and roof
Construction	**Unit 2** Process Critical Utilities & Total Tool Installation Solutions (PCU & TTI) Re-equips a live semiconductor fab. Standalone On the EPC or EPCM model	MEP (Mechanical, Electrical and Plumbing)	Cleanroom envelope
Testing and commissioning			Architectural finishes
Maintenance	**Unit 3** EFM (Energy Facility Management) Runs and maintains it. Cooling as a Service	Process utilities The utilities a production tool consumes	Air conditioning and ventilation (HVAC, ACMV), fan filter units
			Process cooling water (PCW)
			Chiller plant and district cooling
			Fire protection
			Electrical, high and low voltage
			Plumbing and drainage
			Clean dry air (CDA)
			Specialty gases and chemical delivery
			Ultrapure water (UPW)
			Process vacuum (PV) and exhaust
			Waste treatment
			Gas, chemical, water and exhaust connections to each tool


filter units, Process cooling water (PCW), Chiller plant and district cooling,
Fire protection, Electrical, high and low voltage, Plumbing and drainage.
Tools hookup
Energy Performance Contracting
Build Operate Transfer
Tools hookup
to each tool
Release to production
MEP picked

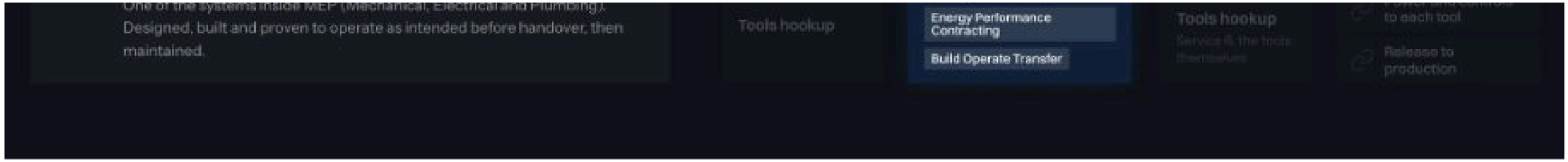
The IAQ facility map



+ − ▶ Play the tour

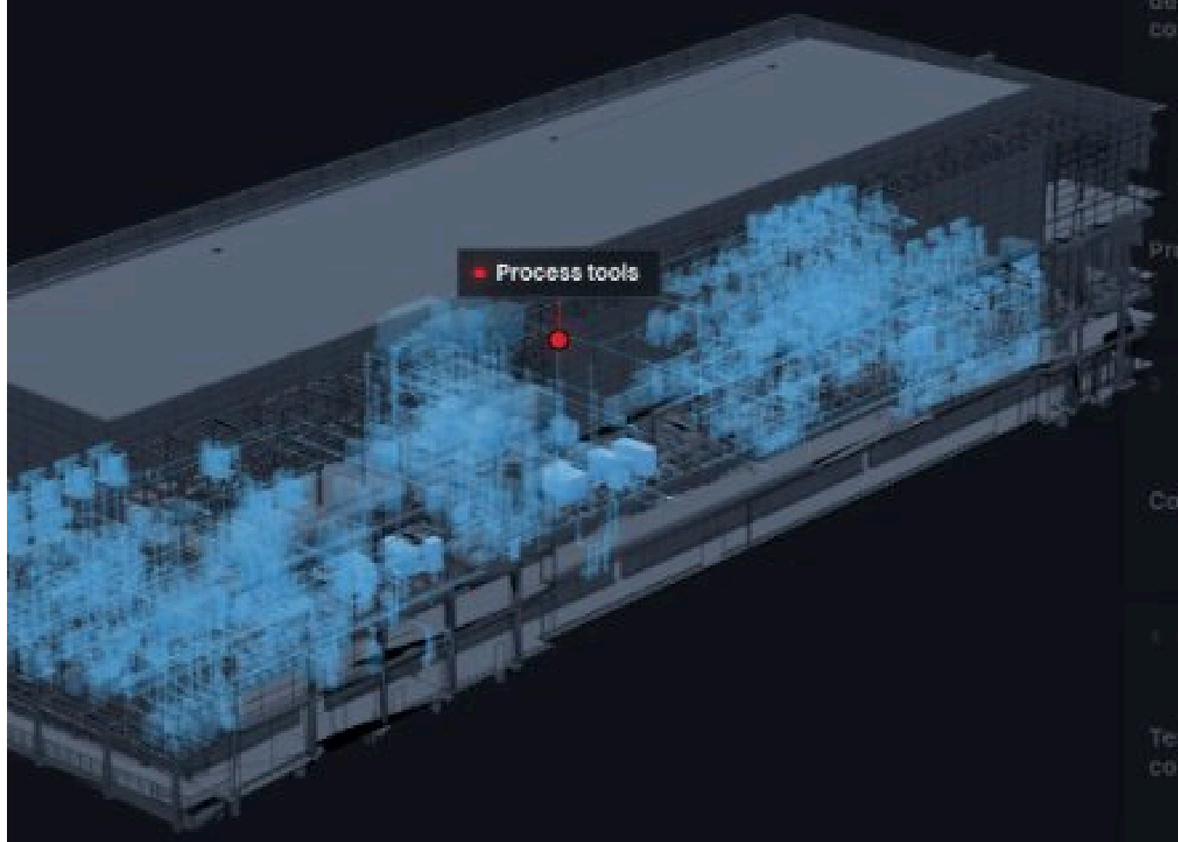
Fire protection is a system, not a service. It is built in **MEP** (Mechanical, Electrical and Plumbing) work, delivered by **EPC** (Engineering, Procurement and Construction), **PCU & TTI** (Process Critical Utilities & Total Tool Installation) **and EFM** (Energy Facility Management).

Service	Business unit	Work	System
Engineering design and consultation	**Unit 1** EPC (Engineering, Procurement and Construction) Builds the facility. EPCC · EPCM	CSA (Civil, Structural and Architectural)	Piling and foundations
Procurement			Structural frame and roof
Construction	**Unit 2** Process Critical Utilities & Total Tool Installation Solutions (PCU & TTI) Re-equips a live semiconductor fab. Standalone · On the EPC or EPCM model	MEP (Mechanical, Electrical and Plumbing)	Cleanroom envelope
Testing and commissioning			Architectural finishes
Maintenance	**Unit 3** EFM (Energy Facility Management) Runs and maintains it. Cooling as a Service	Process utilities The utilities a production tool consumes	Air conditioning and ventilation (HVAC, ACMV), fan filter units
			Process cooling water (PCW)
			Chiller plant and district cooling
			Fire protection
			Electrical, high and low voltage
			Plumbing and drainage
			Clean dry air (CDA)
			Specialty gases and chemical delivery
			Ultrapure water (UPW)
			Process vacuum (PV) and exhaust
			Waste treatment
			Gas, chemical, water and exhaust connections to each tool



Fire protection picked

The IAQ facility map



+ − ▶ Play the tour

Tools hookup: production tools connected into the facility and released to production. Carried by **PCU & TTI** (Process Critical Utilities & Total Tool Installation). Any one service can be bought on its own.

Service

- Engineering design and consultation
- Procurement
- Construction
- Testing and commissioning
- Maintenance

Business unit

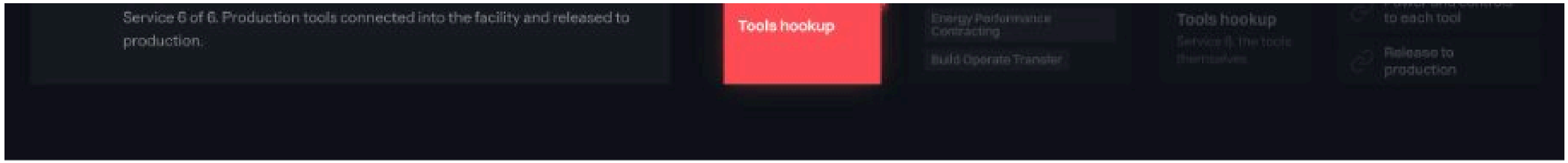
- EPC (Engineering, Procurement and Construction) — Builds the facility
- EPCC
- EPCM
- Unit 2 — **Process Critical Utilities & Total Tool Installation Solutions** (PCU & TTI) — Re-equips a live semiconductor fab.
  - Standalone
  - On the EPC or EPCM model
- Unit 3 — EFM (Energy Facility Management) — Runs and maintains it
- Cooling as a Service

Work

- CSA (Civil Structural and Architectural)
- MEP (Mechanical, Electrical and Plumbing)
- Process utilities — The utilities a production tool consumes

System

- Piling and foundations
- Structural frame and roof
- Cleanroom envelope
- Architectural finishes
- Air conditioning and ventilation (HVAC, ACMV), fan filter units
- Process cooling water (PCW)
- Chiller plant and district cooling
- Fire protection
- Electrical, high and low voltage
- Plumbing and drainage
- Clean dry air (CDA)
- Specialty gases and chemical delivery
- Ultrapure water (UPW)
- Process vacuum (PV) and exhaust
- Waste treatment
- Gas, chemical, water and exhaust connections to each tool



Tools hookup picked

3D archive.
The assembly the home page ran until 26 September 2026.

Brand Method's own scroll-driven build of IAQ's model: six stages, one accountable rail, the finish. Replaced on the home page by the developer's Scroll to build 3D (delivery "3D ONLY (4)", 26 September 2026), which assembles the same facility in 25 beats under IAQ's service headings and hands off to a first-person walkthrough. Kept here in full.



The facility,
layer by layer

9

Overall

Review of all eight systems · envelope closes · plant lands · site handover

9 / 9 · OVERALL

Civil & Structural
Architectural
Process Utilities
Fire Protection
Air-conditioning
HT & LV Electrical
Cleanroom System
Tools Hookup

Skip the build ↓

5









